

13 APPENDIX

13.1 Summary

This chapter provides electric-related information based on CCW-RV-214W* model device for reference.

Your device may be different from the one described in this chapter.

<Contents>

- Board location of the electrical unit
- Overall wiring diagram and block wiring diagram

<Purpose>

- To provide reference information for repairing the device or replacing parts.

<Intended reader>

- Maintenance engineers
- Ishida service engineers

13.2 Electrical Unit Layout Drawing

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

13.2.1 Layout Drawing for CCW-RV-214W-S/20, 30

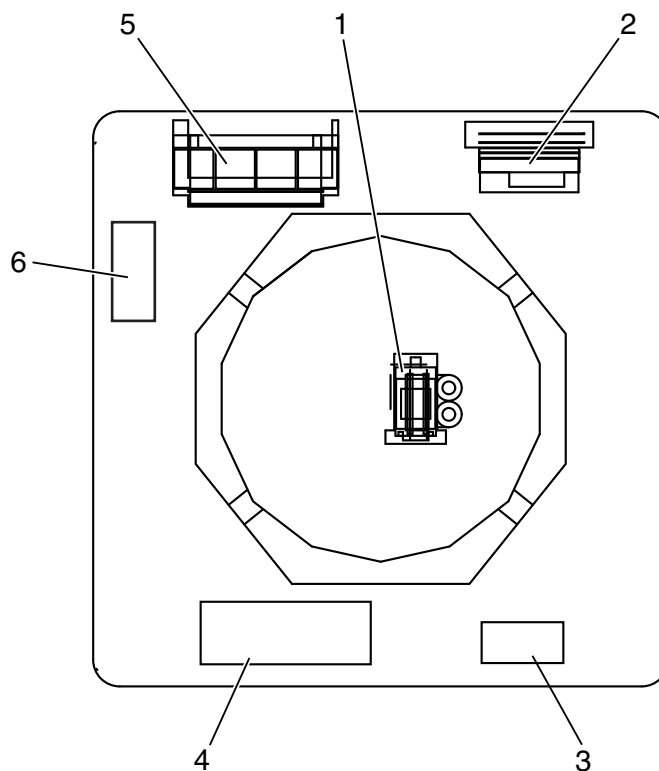


Fig. 13-1

Table 13-1

No.	Name
1	Main electrical unit (HUB board, MPS board, FDRV board, FDC board)
2	CAL unit (ADC board, WCU board, DMU board)
3	RELAY unit (EXC board, RELAY board)
4	PS-0 unit
5	PS-2 unit
6	PS-FEEDER unit

13.2.2 Layout Drawing for CCW-RV-214W-S/15

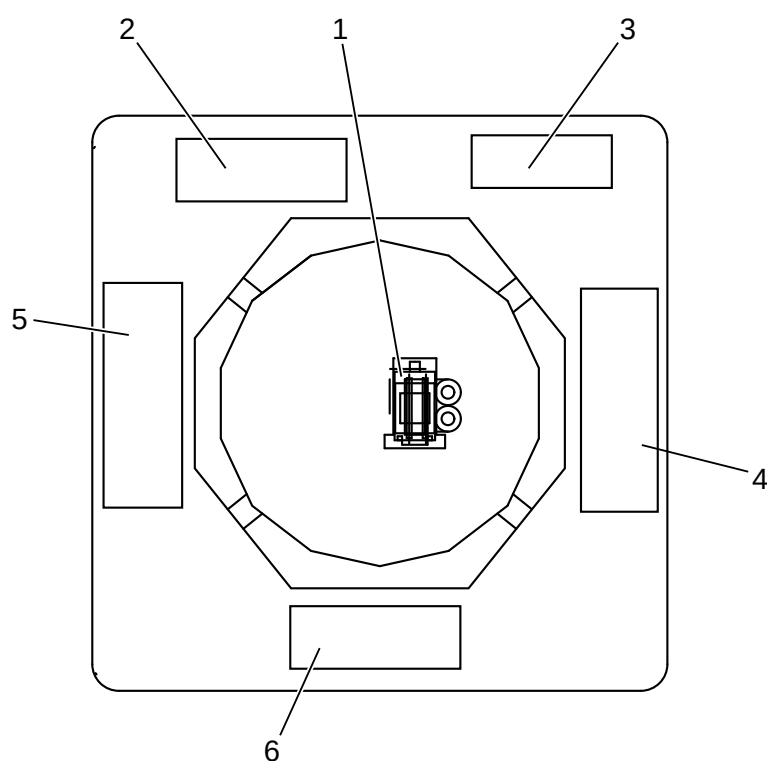


Fig. 13-2

Table 13-2

No.	Name
1	Main electrical unit (HUB board, MPS board, FDRV board, FDC board)
2	CAL unit (ADC board, WCU board, DMU board)
3	RELAY unit (EXC board, RELAY board)
4	PS-0 unit
5	PS-2 unit
6	PS-FEEDER unit

13.2.3 Layout Drawing for CCW-RV-214W-S/50, 70

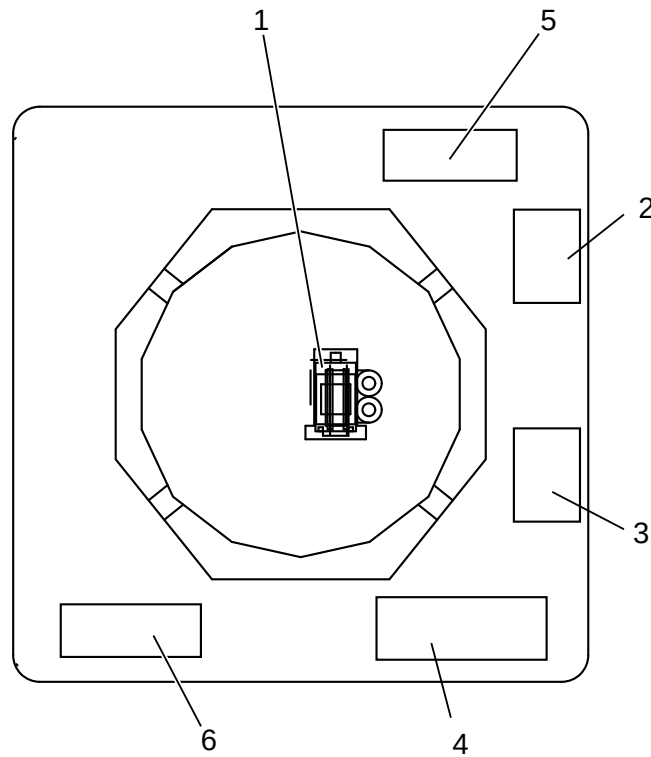


Fig. 13-3

Table 13-3

No.	Name
1	Main electrical unit (HUB board, MPS board, FDRV board, FDC board)
2	CAL unit (ADC board, WCU board, DMU board)
3	RELAY unit (EXC board, RELAY board)
4	PS-0 unit
5	PS-2 unit
6	PS-FEEDER unit

13.3 Electrical Unit Layout Drawing (CCW-RV-214W-S/05, 08)

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

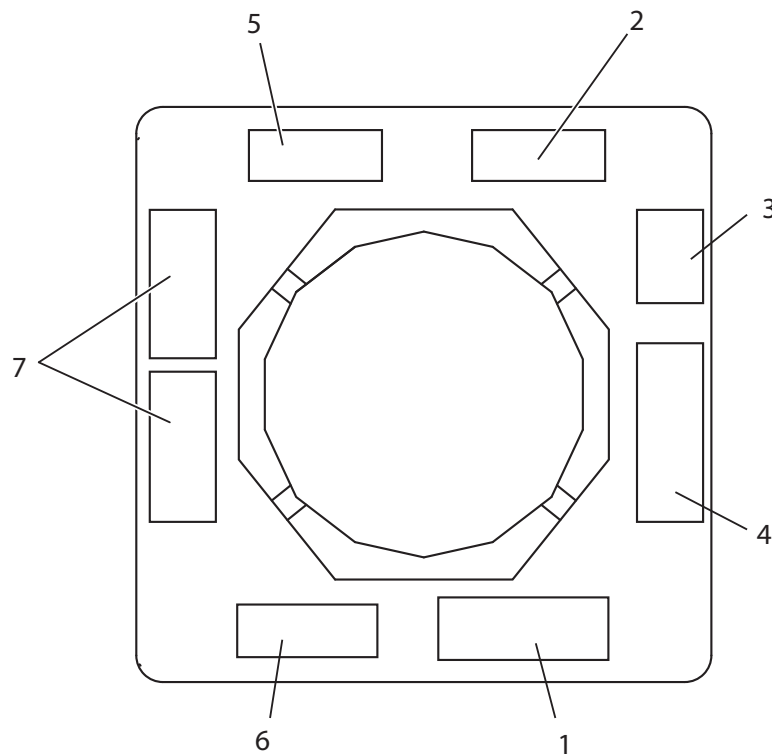


Fig. 13-4

Table 13-4

No.	Name
1	Main electrical unit (FDRV board, FDC board)
2	CAL unit (ADC board, WCU board, DMU board)
3	RELAY unit (RELAY board)
4	PS-0 unit
5	PS-2 unit
6	PS-FEEDER unit
7	DUC unit (DUC board)

13.4 Overall Block Diagram

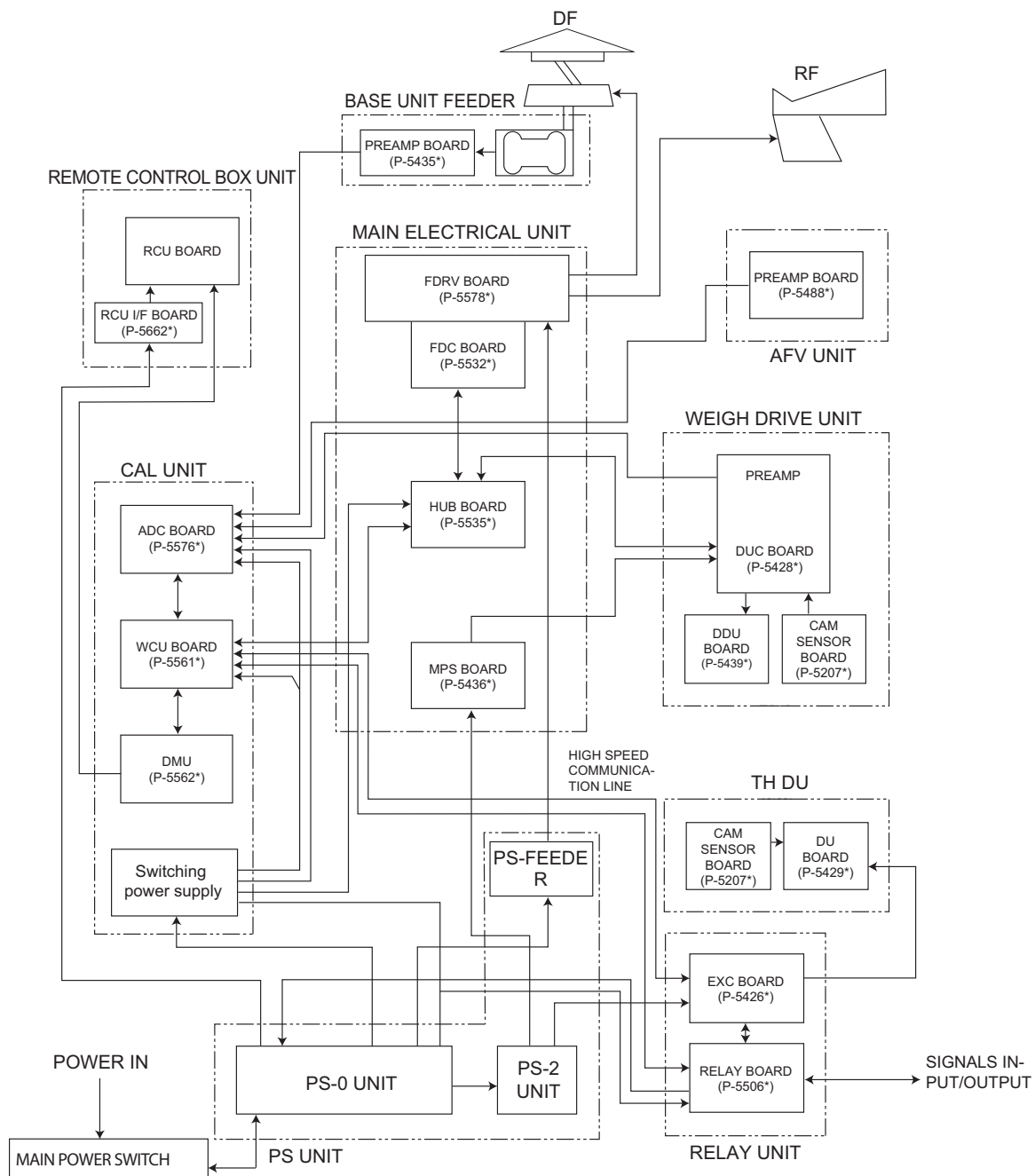


Fig. 13-5

NOTE

- The drawing is applicable to CCW-RV-214W-S/15, 20, 30, 50, 70.

13.5 Overall Block Diagram (CCW-RV-214W-S/05, 08)

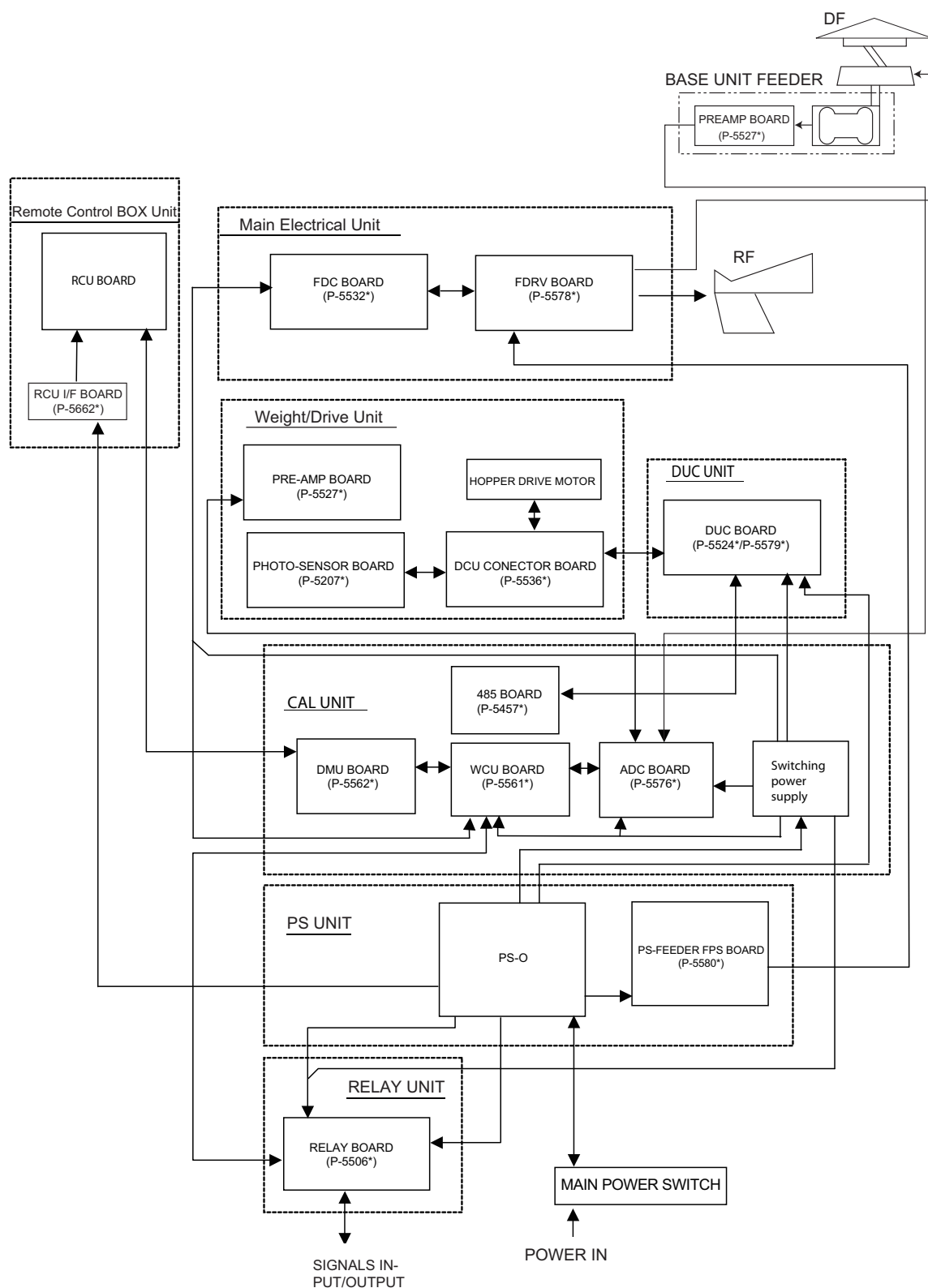


Fig. 13-6

13.6 Remote Control BOX Unit

! DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

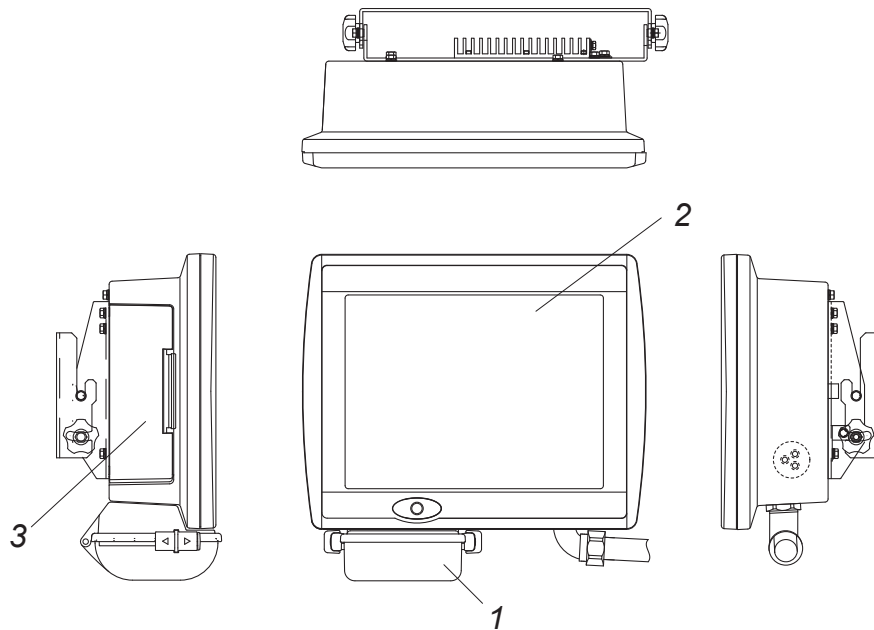


Fig. 13-7

Table 13-5

No.	Name
1	Printer unit
2	Touch panel display
3	USB insertion slot

13.6.1 RCU Board

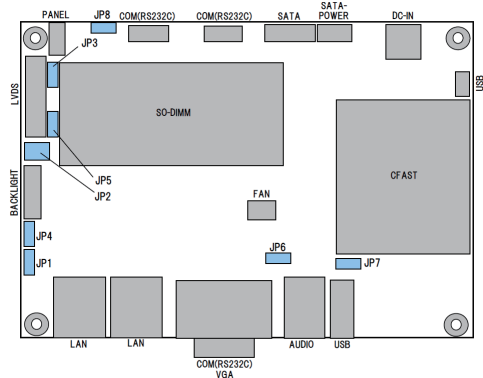


Fig. 13-8

Functions of the board

- 1. Input control and display control of the touch panel.
- 2. Printer control and statistic calculation processing.
- 3. Communication with the CAL unit (DMU board) via a high-speed communication line.
- 4. Input and output control of the USB.

<Jumper connection and associated function>

JP1

Table 13-6

	JP1	Function
NC	1-2	—
NC	2-3	—

JP2

Table 13-7

LVDS_VCC	JP2	Function
3.3 V	1-3 2-4 (Default)	—
5 V	3-5 4-6	—

JP3

Table 13-8

SELLVDS	JP3	Function
High	1-2	LVDS connector - pin#9 (pin No.9) output signal setting High level is LVDS_VCC.
Low	2-3	

JP4**Table 13-9**

ENABKL	JP4	Function
3.3 V	1-2 (Default)	BACKLIGHT connector - pin#1 (pin No.1) output signal setting Setting high level
5 V	2-3	

JP5**Table 13-10**

LVDS0_D3+	JP5	Function
1.8 V	1-2 (Default)	LVDS connector - pin#29 (pin No.29) output signal setting
3.3 V	2-3	

JP6**Table 13-11**

Clear CMOS	JP6	Function
Normal	1-2 (Default)	—
Clear	CMOS 2-3	—

JP7**Table 13-12**

Supply voltage for CAST	JP7	Function
3.3 V	1-2 (Default)	—
5 V	2-3	—

JP8**Table 13-13**

	JP8	Function
Hardware Auto Power On	1-2	Enables "power-on start-up" independently of setting of "Restore AC Power Loss of BIOS".
NC	2-3 (Default)	—

13.6.2 PRN Board

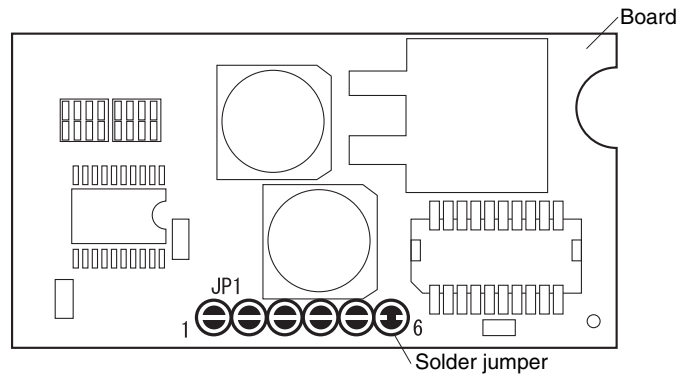


Fig. 13-9

Functions of the board

1. Control of the thermal printer (Printing or paper feeding).

Setting of board jumper

Set the data input format using the solder jumpers JP1 to JP6.
Set JP1 to JP5 to open, and JP6 to short.

13.6.3 TP-I/F Board (P-5661*)

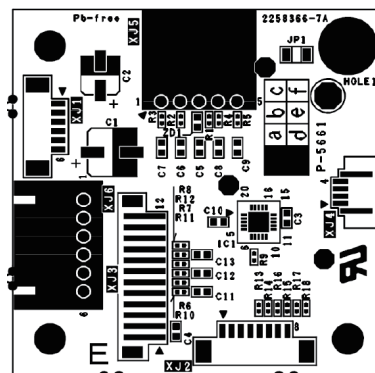


Fig. 13-10

Functions of the board

1. Control of the touch panel.

13.6.4 RCU I/F Board (P-5662*)

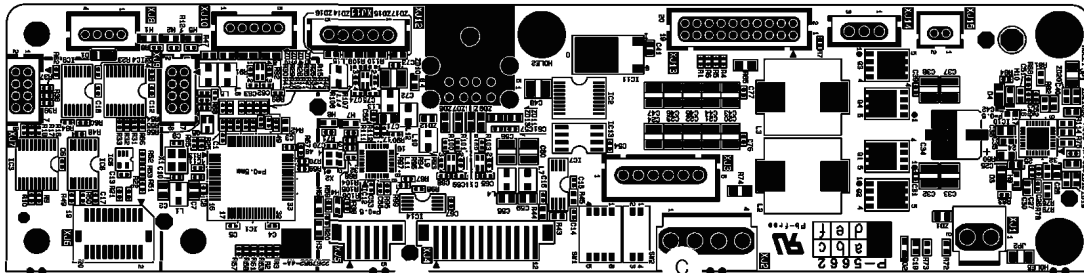


Fig. 13-11

Functions of the board

1. Power supply to RCU board (+12 V).
2. Trunk circuit for USB communication with PRN board or TP-I/F board.

<DIP-SW settings>

SW1

Table 13-14

No.	Function	Default Setting
1	Not used	OFF
2	Not used	OFF
3	For reboot with 45sec delay	OFF
4	Not used	OFF

SW2

Table 13-15

No.	Function	Default Setting
1	Printer correspondent (ON:XJ6)	ON
2	Touch panel setting	OFF

13.7 Main Electrical Unit

**DANGER**

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

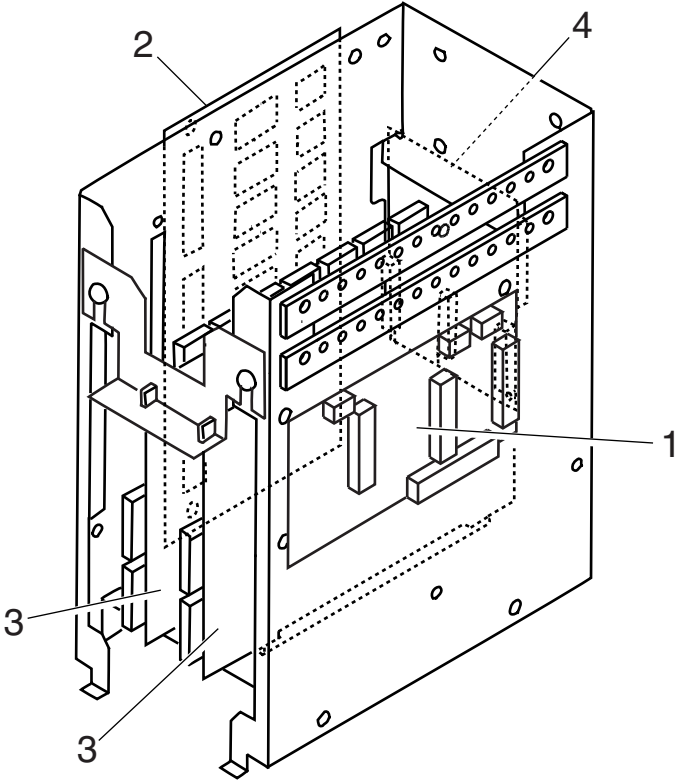


Fig. 13-12

Table 13-16

No.	Name
1	HUB board (P-5535*)
2	MPS board (P-5436*)
3	FDRV board (P-5578*)
4	FDC board (P-5532*)

NOTE

- The drawing is applicable to CCW-RV-214W-S/15, 20, 30, 50, 70.

13.7.1 HUB Board (P-5535*)

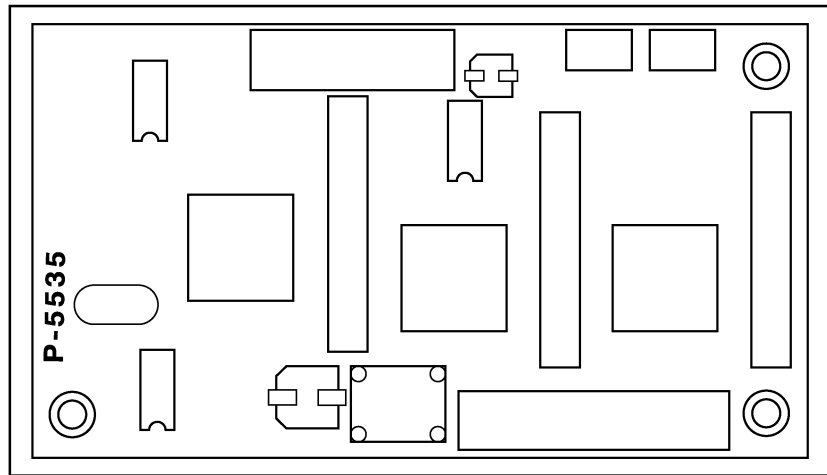


Fig. 13-13

Functions of the board

1. Relay and distribution of communication between WCU board, DUC board and FDC board via a high-speed communication line.

NOTE

- Check the suffix number when replacing the board.

13.7.2 MPS Board (P-5436*)

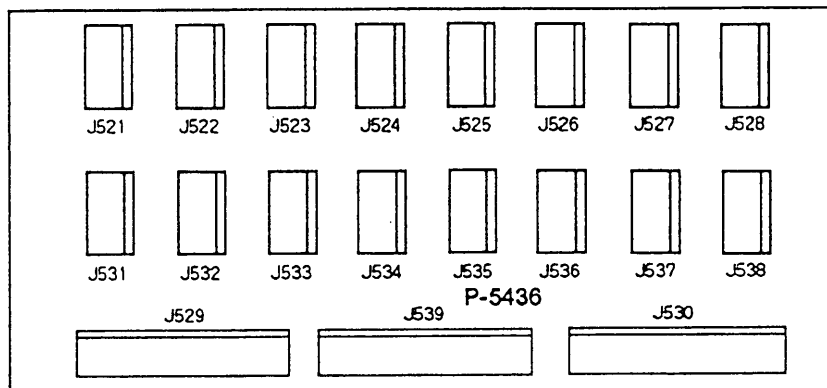


Fig. 13-14

Functions of the board

1. Relay and distribution of 39 VDC power supply from PS-2 unit to DUC board.

13.7.3 FDRV Board (P-5578*)

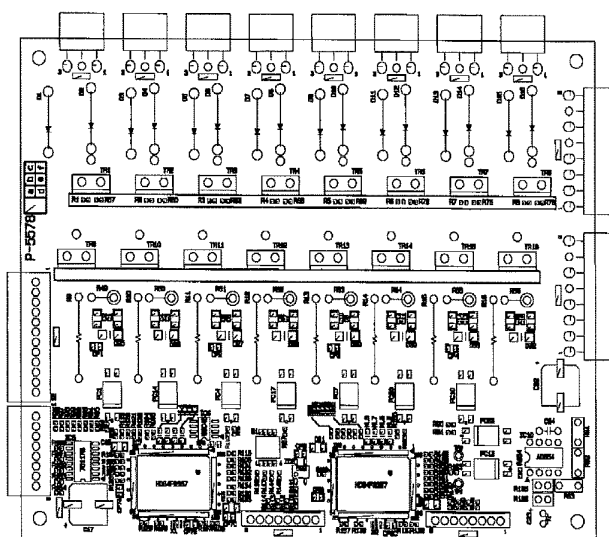


Fig. 13-15

Functions of the board

1. Drive control of the radial feeder.
2. Relay and distribution of power supply from PS feeder unit to radial feeder.

13.7.4 FDC Board (P-5532*)

NOTE

- When replacing the board, set the DIP switch to the setting as it was before the replacement.

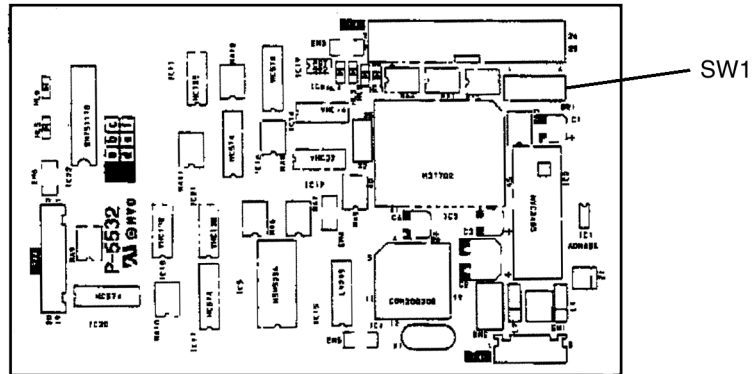


Fig. 13-16

Functions of the board

- Drive control of the radial feeder.

Setting of the board switches

Table 13-17

DIP SW	Setting
SW 1-1	OFF
SW 1-2	OFF
SW 1-3	OFF
SW 1-4	OFF

13.8 Main Electrical Unit (CCW-RV-214W-S/05, 08)

**DANGER**

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

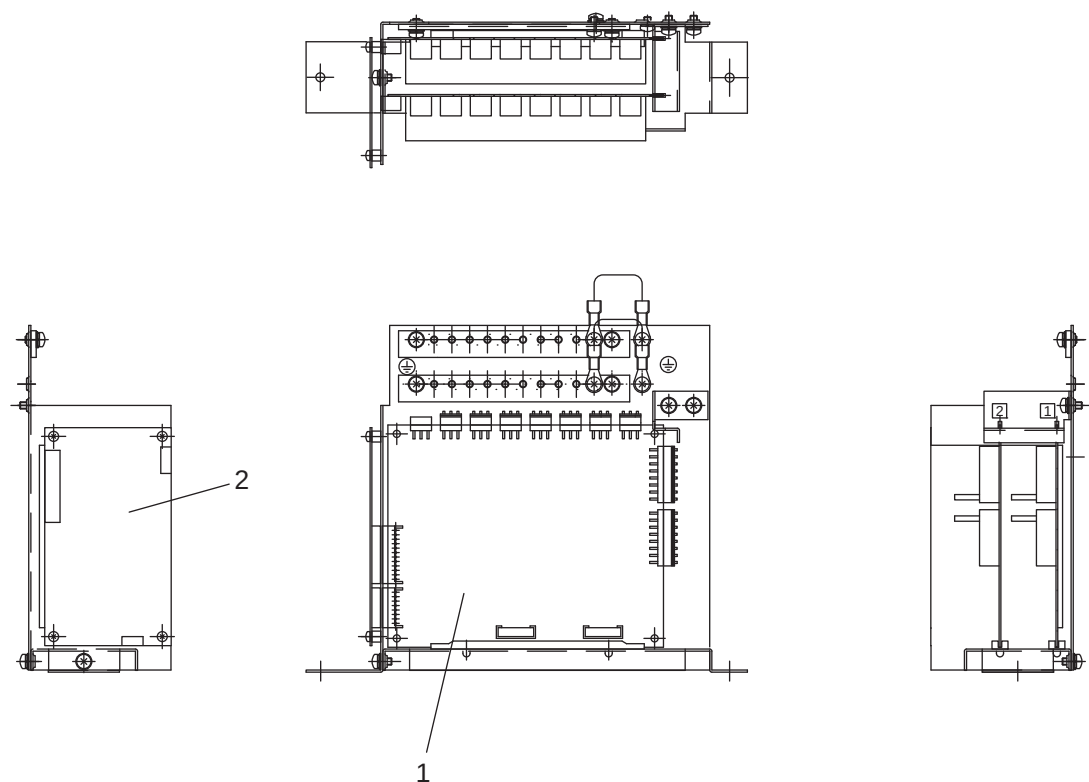


Fig. 13-17

Table 13-18

No.	Name
1	FDRV board (P-5578*)
2	FDC board (P-5532*)

NOTE

- The figure is applicable to CCW-RV-214W-S/05, 08.

Setting of the board switches

Table 13-19

DIP SW	Setting
SW 1-1	OFF
SW 1-2	OFF
SW 1-3	OFF
SW 1-4	OFF

13.9 Base Unit Feeder

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

13.9.1 Preamp Board (P-5435*)

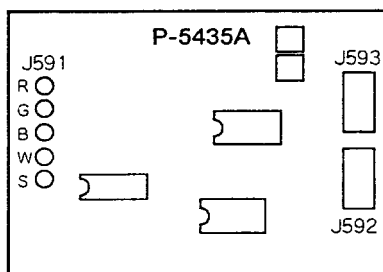


Fig. 13-20

Functions of the board

1. Amplification of the dispersion feeder load cell output.

NOTE

- Check the suffix number when replacing the board.

13.10 AFV Unit

! DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

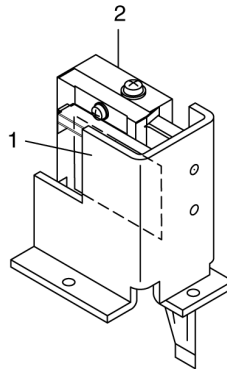


Fig. 13-21

Table 13-20

No.	Name
1	PRE-AMP Board
2	Load cell (BAS-0.25L)

13.10.1 AFV Preamp Board (P-5488*)

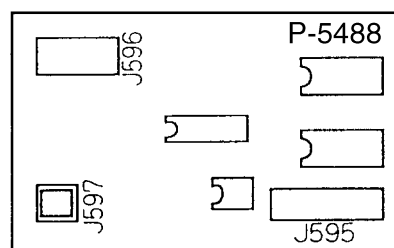


Fig. 13-22

Functions of the board

1. Amplification of the AFV load cell output.

13.11 Relay Unit

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

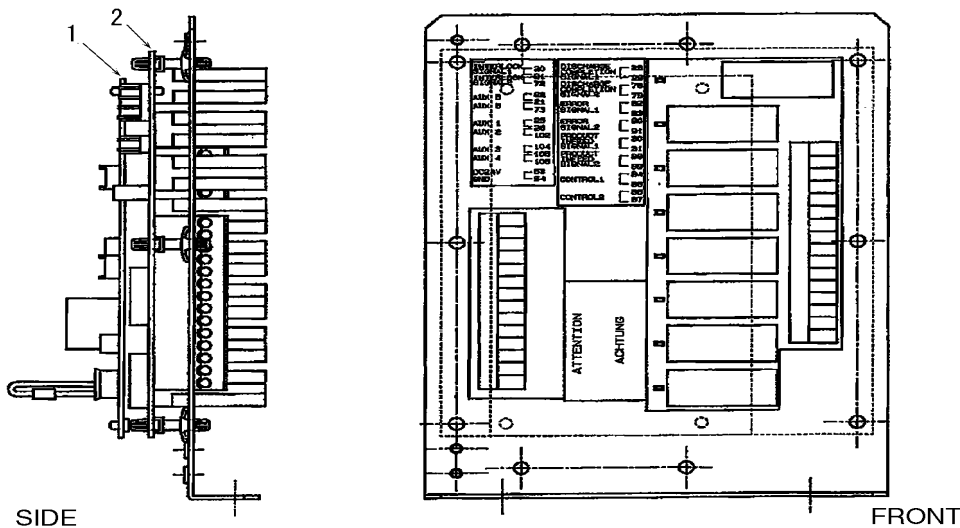


Fig. 13-23

Table 13-21

No.	Name
1	EXC board (P-5426*)
2	RELAY board (P-5506*)

13.11.1 EXC Board (P-5426*)

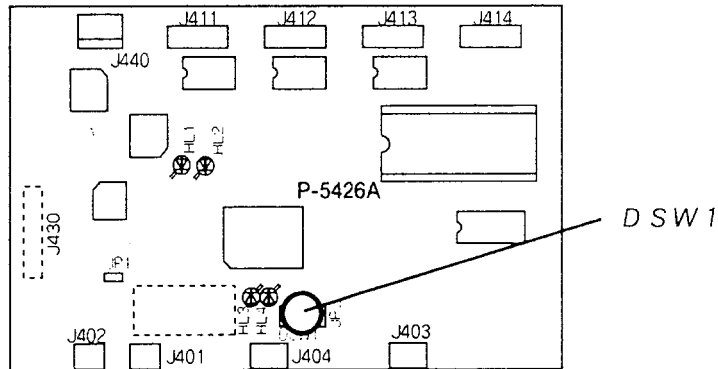


Fig. 13-24

* In most instances, set all DSW1 switches to OFF.

Functions of the board

1. Communication with the CAL unit.
2. Control of the timing hopper (Optional).

NOTE

- The CCW-RV-214W-S/05, 08 type has no EXC board mounted.

13.11.2 Relay Board (P-5506*)

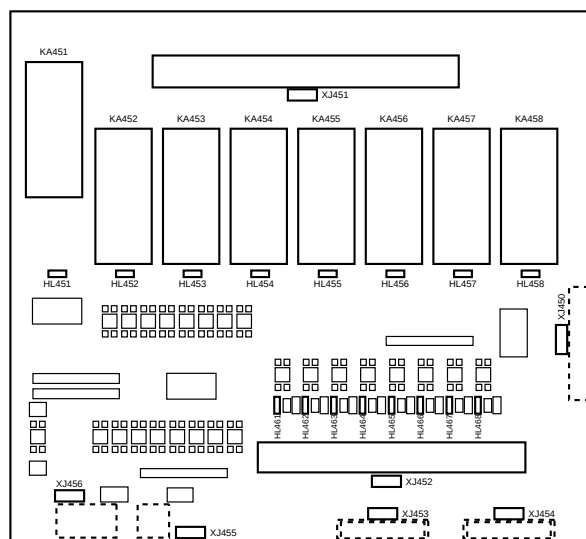


Fig. 13-25

Functions of the board

1. Input and input monitoring of the interlock signal from the packer.
2. Output and output monitoring of the discharge completion signal to the packer.
3. Output and output monitoring of the error signal.
4. Output and output monitoring of the infeed control signal to the infeeder.
5. Input and input monitoring of the optional signal.
6. Output and output monitoring of the optional signal.

Functions of monitoring LED**Table 13-22**

LED	Function	LED	Function
HL 451	Discharge completion signal 1	HL 461	Interlock signal 1
HL 452	Discharge completion signal 2	HL 462	Interlock signal 2
HL 453	Error signal 1	HL 463	Input signal 5
HL 454	Error signal 2	HL 464	Input signal 6
HL 455	Infeed control signal 1	HL 465	Input signal 1
HL 456	Infeed control signal 2	HL 466	Input signal 2
HL 457	Control 1	HL 467	Input signal 3
HL 458	Control 2	HL 468	Input signal 4

13.12 PS-0 Unit

! DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

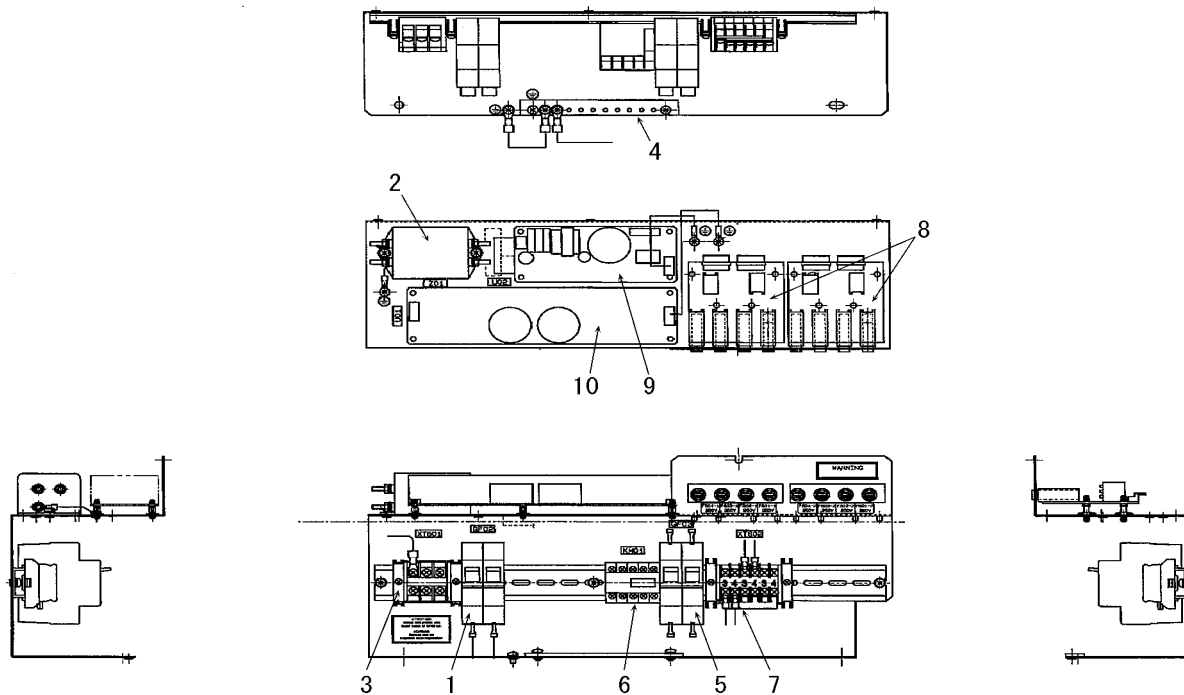


Fig. 13-26

Table 13-23

No.	Name
1	Circuit breaker
2	Noise filter
3	Terminal block
4	Grounding block
5	Circuit breaker
6	Electromagnetic contactor
7	Terminal block
8	AC FUSE board (P-5507*)
9	Switching power supply for relay unit (24 VDC)
10	Switching power supply for remote control BOX (24 VDC)

Functions of the unit

- 1. Detects overcurrent and shuts the power off.
- 2. Removes noise from the power source.
- 3. DC power supply to the remote control BOX unit and relay unit.
- 4. AC power supply to the main electrical unit, PS-FEEDER unit and CAL unit.

13.13 PS-2 Unit



DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

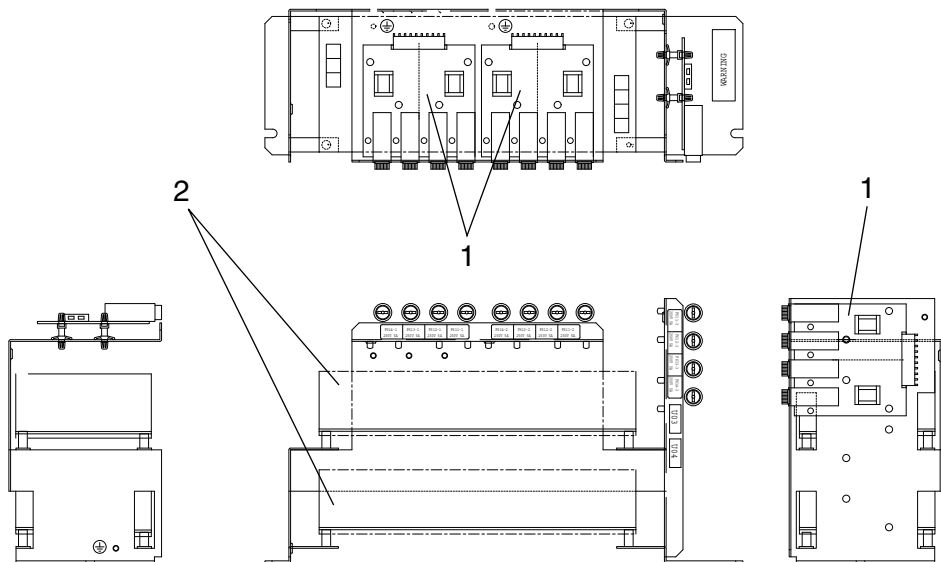


Fig. 13-27

Table 13-24

No.	Name
1	DC FUSE board (P-5508*)
2	39 VDC switching power supply

Functions of the unit

- 1. Power supply to the motor of the weigh drive unit.

NOTE

- The number of the fuse boards may differ according to the number of heads in the machine.
- The CCW-RV-214W-S/15, 20, 30, 50, 70 type uses 39 VDC power supply.
- The CCW-RV-214W-S/05, 08 type uses 24 VDC power supply.

13.14 Weigh Drive Unit

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

NOTE

- The drawing is applicable to CCW-RV-214W-S/15, 20, 30, 50, 70.

13.14.1 DUC Board (P-5428*)

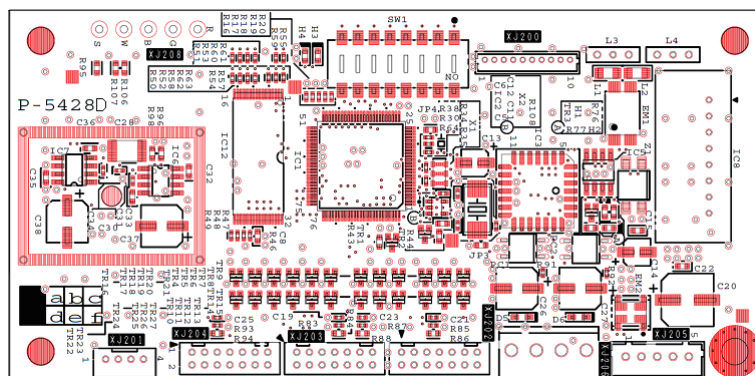


Fig. 13-28

Functions of the board

1. Communication with the WCU board by a high-speed communication line via HUB board.
2. Open/close control of the pool hopper and weigh hopper.
3. Amplification of the load cell output by preamp circuit.

NOTE

- When replacing the weigh drive unit and DUC board, perform the head setting by DIP-SW.

DIP-SW settings**Table 13-25**

Head No.	SW1-1	SW1-2	SW1-3	SW1-4	SW1-5	SW1-6	SW1-7	SW1-8
1	ON	OFF	OFF	OFF	OFF	OFF	OFF	OFF
2	OFF	ON	OFF	OFF	OFF	OFF	OFF	OFF
3	ON	ON	OFF	OFF	OFF	OFF	OFF	OFF
4	OFF	OFF	ON	OFF	OFF	OFF	OFF	OFF
5	ON	OFF	ON	OFF	OFF	OFF	OFF	OFF
6	OFF	ON	ON	OFF	OFF	OFF	OFF	OFF
7	ON	ON	ON	OFF	OFF	OFF	OFF	OFF
8	OFF	OFF	OFF	ON	OFF	OFF	OFF	OFF
9	ON	OFF	OFF	ON	OFF	OFF	OFF	OFF
10	OFF	ON	OFF	ON	OFF	OFF	OFF	OFF
11	ON	ON	OFF	ON	OFF	OFF	OFF	OFF
12	OFF	OFF	ON	ON	OFF	OFF	OFF	OFF
13	ON	OFF	ON	ON	OFF	OFF	OFF	OFF
14	OFF	ON	ON	ON	OFF	OFF	OFF	OFF
15	ON	ON	ON	ON	OFF	OFF	OFF	OFF
16	OFF	OFF	OFF	OFF	ON	OFF	OFF	OFF
17	ON	OFF	OFF	OFF	ON	OFF	OFF	OFF
18	OFF	ON	OFF	OFF	ON	OFF	OFF	OFF
19	ON	ON	OFF	OFF	ON	OFF	OFF	OFF
20	OFF	OFF	ON	OFF	ON	OFF	OFF	OFF
21	ON	OFF	ON	OFF	ON	OFF	OFF	OFF
22	OFF	ON	ON	OFF	ON	OFF	OFF	OFF
23	ON	ON	ON	OFF	ON	OFF	OFF	OFF
24	OFF	OFF	OFF	ON	ON	OFF	OFF	OFF

13.14.2 DDU Board (P-5439*)

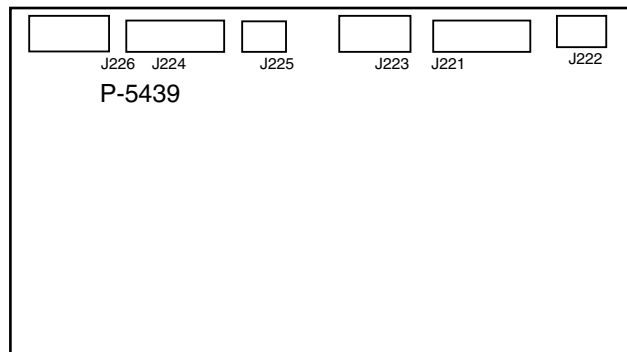


Fig. 13-29

Functions of the board

1. Driving the motor for opening/closing the pool hopper and the weigh hopper.

13.14.3 Cam Sensor Board (P-5207*)

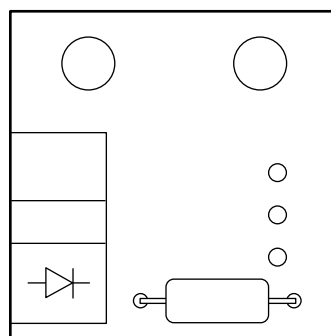


Fig. 13-30

Functions of the board

1. Detects the stop position of the pool hopper and the weigh hopper using the cam with slit, and sends the signal to the DUC board.

13.15 Weigh Drive Unit (CCW-RV-214W-S/05, 08)

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

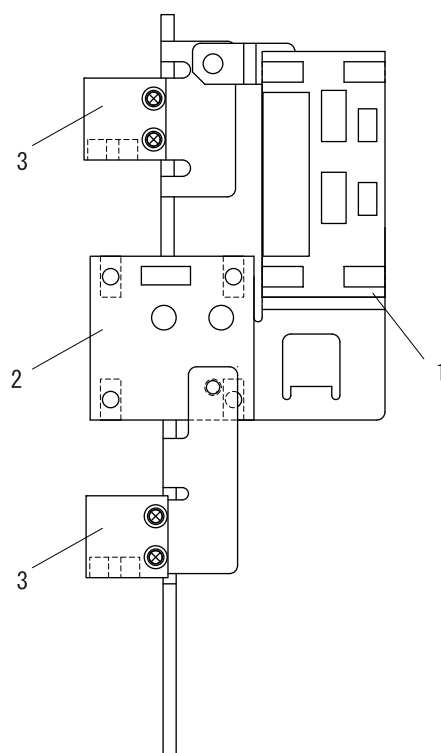


Fig. 13-31

Table 13-26

No.	Name
1	DUC connector board (P-5536*)
2	Preamp board (P-5527*)
3	Photo sensor board (P-5207*)

13.15.1 DUC Connector Board (P-5536*)

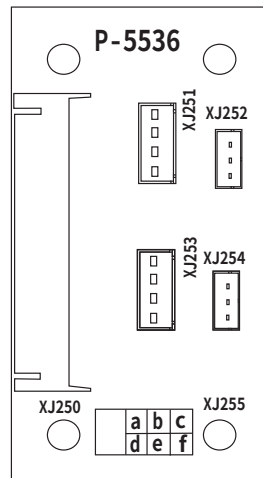


Fig. 13-32

Functions of the board

1. Relay of pool hopper (PH), weigh hopper (WH), timing hopper (TH) and drive signals from the DUC board and of sensor signals from the photo sensor.

13.15.2 Preamp Board (P-5527*)

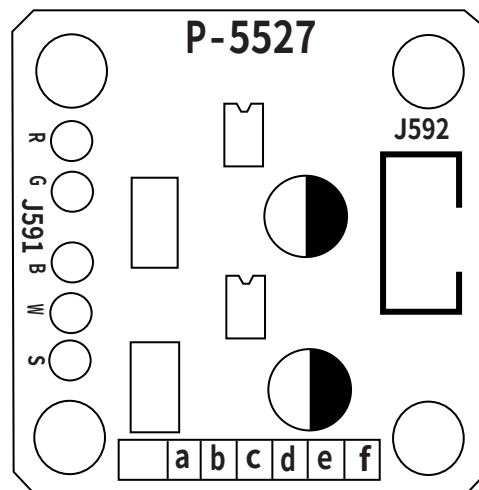


Fig. 13-33

Functions of the board

1. Amplification of the load cell output by preamp circuit.

13.15.3 Photo Sensor Board (P-5207*)

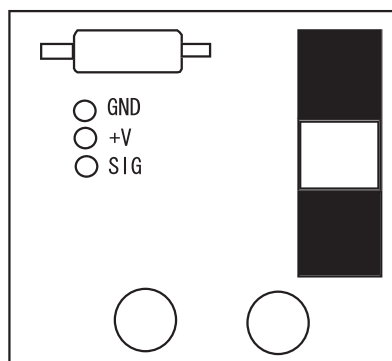


Fig. 13-34

Functions of the board

1. Detection of stop positions of the stepping motor for opening/closing the pool hopper (PH) and weigh hopper (WH).
2. Detection of stop positions of the stepping motor for opening/closing the timing hopper (TH).

13.16 CAL Unit

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

NOTE

- The drawing is applicable to CCW-RV-214W-S/15, 20, 30, 50, 70.

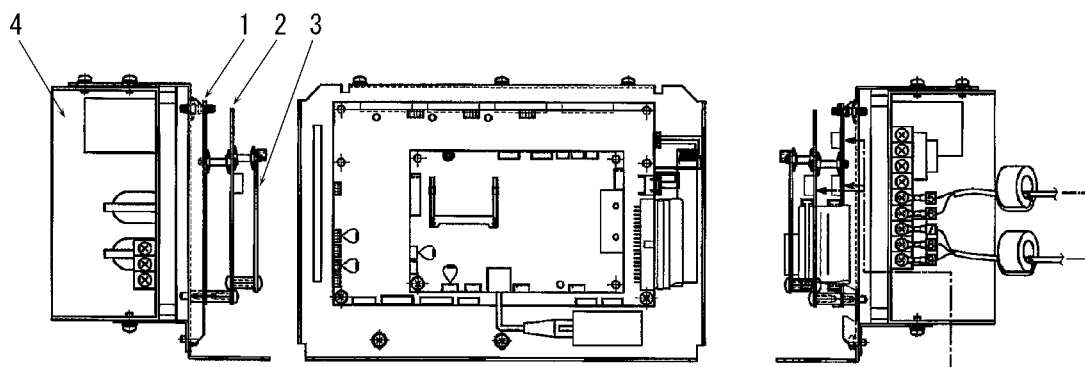
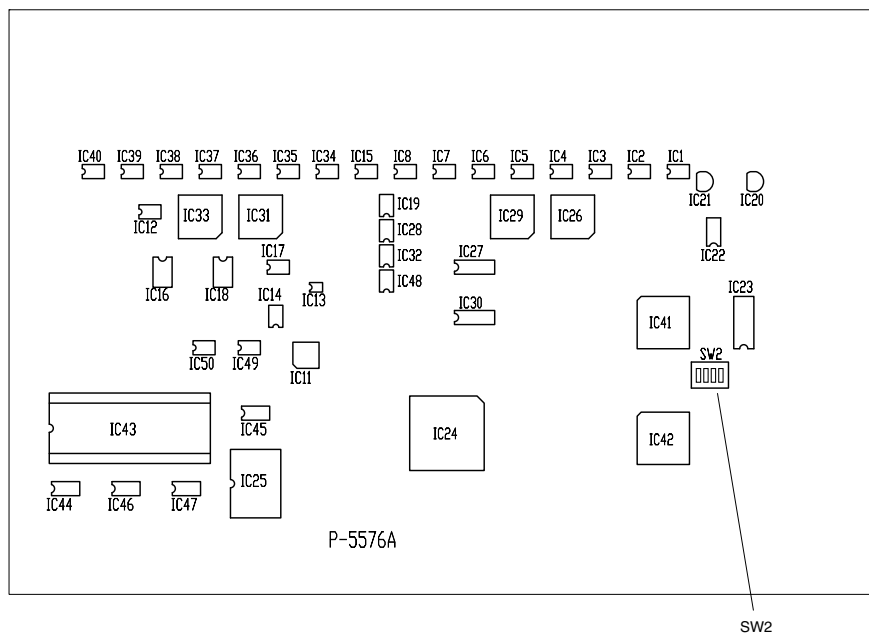


Fig. 13-35

Table 13-27

No.	Name
1	ADC board (P-5576*)
2	WCU board (P-5561*)
3	DMU board (P-5562*)
4	Switching power supply for CAL unit (5 VDC, ± 15 V)

13.16.1 ADC Board (P-5576*)

**Fig. 13-36**

Functions of the board

1. Converts the analog weight data from each weigh unit (load cell) to digital weight data, and sends it to the WCU board.

DIP-SW settings

Table 13-28

DIP SW	Function	Default Setting
SW2-1	ON (Starts DSP from PROM) / OFF (Starts DSP from FLASH)	ON
SW2-2	Not used	OFF
SW2-3	For debug	OFF
SW2-4	For debug	OFF

13.16.2 WCU Board (P-5561*)

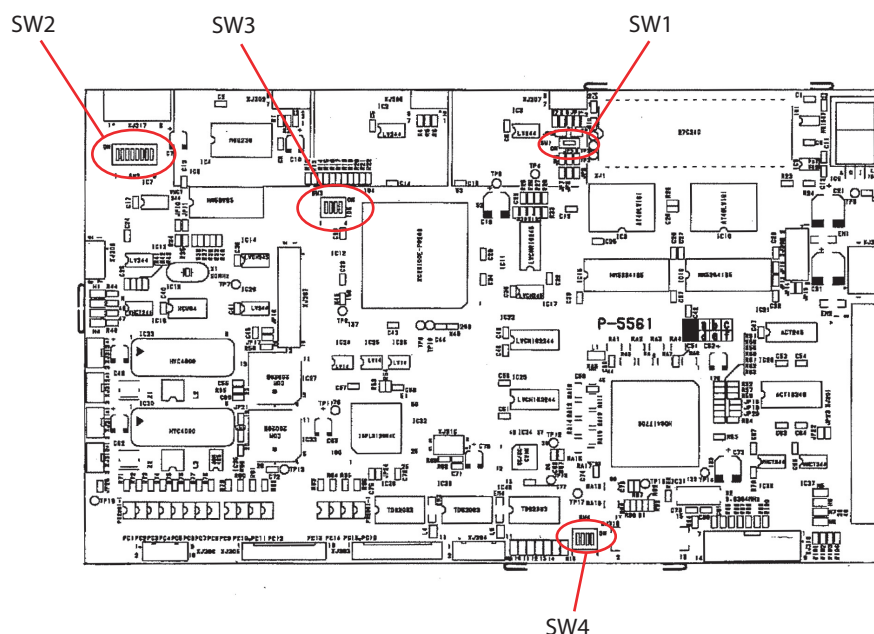


Fig. 13-37

Functions of the board

1. Processing of combination weighing.
2. Sending of control command to each unit based on the combination weighing result.
3. Sending and receiving of various data to and from the DMU board.

DIP-SW settings

Table 13-29

DIP SW	Setting
SW1	OFF
SW2-1 to 8	OFF * Switches may be set to ON depending on the specification. Check the setting when replacing the board.
SW3-1	ON
SW3-2	OFF
SW3-3	ON
SW3-4	OFF
SW4-1	OFF
SW4-2	OFF
SW4-3	ON
SW4-4	OFF

13.16.3 DMU Board (P-5562*)

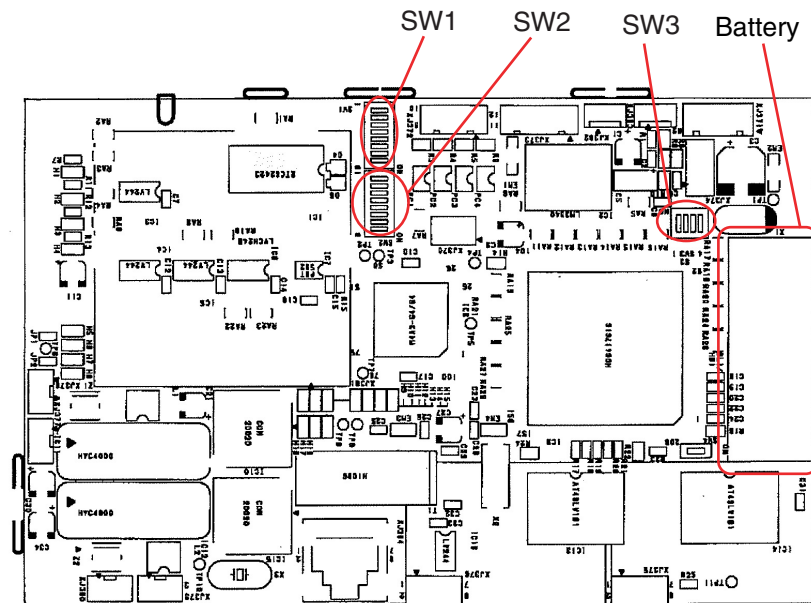


Fig. 13-38

Functions of the board

1. Data processing.
2. Storing of total data and parameters.
3. Communication with the remote control BOX unit (RCU board) by a high-speed communication line.

DIP-SW settings

Table 13-30

DIP SW	Function	Default Setting
SW1-1	ON:Models with two distributed cameras OFF:Models with one distributed camera	OFF
SW1-2	ON:Models with no camera OFF:Models with camera	OFF
SW1-3 to SW1-8	Not used	OFF
SW2-1 to SW2-5	Not used	OFF
SW2-6	ON:Clears SRAM when initializing memory OFF:Does not clear SRAM when initializing memory	OFF
SW2-7	ON:Does not output FIF0 at DMU application OFF:Outputs FIF0 at DMU application	OFF
SW2-8	ON:Does not output boot loader message OFF:Outputs boot loader message	OFF
SW3-1	ON:Boot loader mode OFF:Application execution mode	OFF

Table 13-30

DIP SW	Function	Default Setting
SW3-2	Fix to OFF and should no be turned ON.	OFF
SW3-3	ON:Writing data to flash memory is prohibited. OFF:Writing data to flash memory is allowed.	OFF
SW3-4	Fix to OFF and should no be turned ON.	OFF

 CAUTION

- Refer to "10.3.8 Replacement of Memory Backup Battery" for replacing battery.

13.17 CAL Unit (CCW-RV-214W-S/05, 08)

 DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

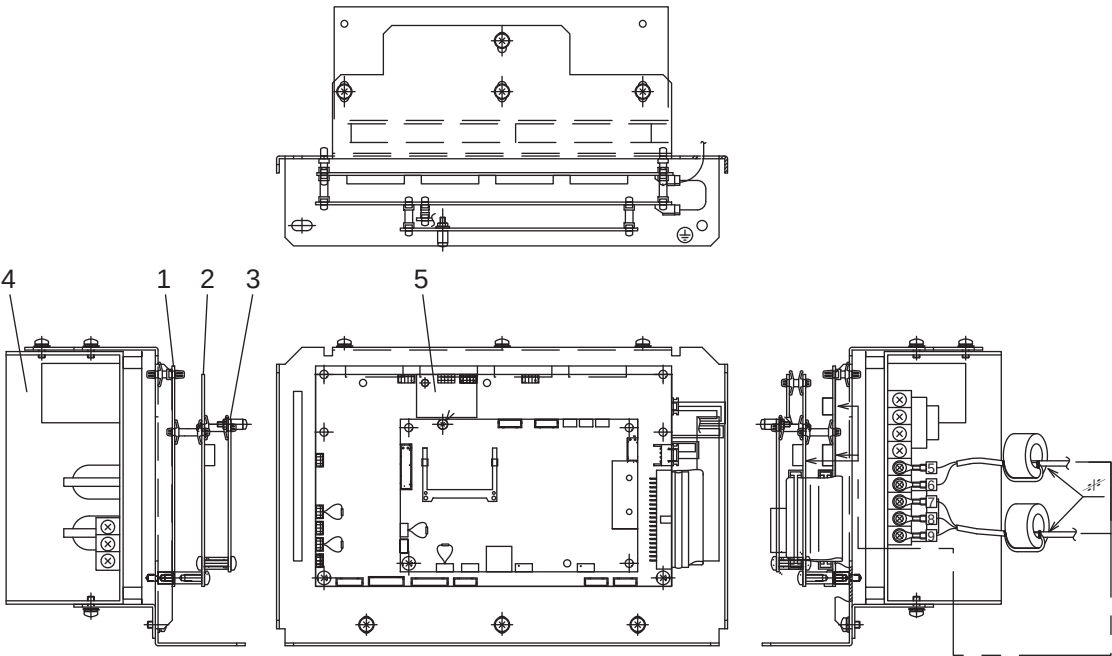


Fig. 13-39

13.17.2 WCU Board (P-5561*)

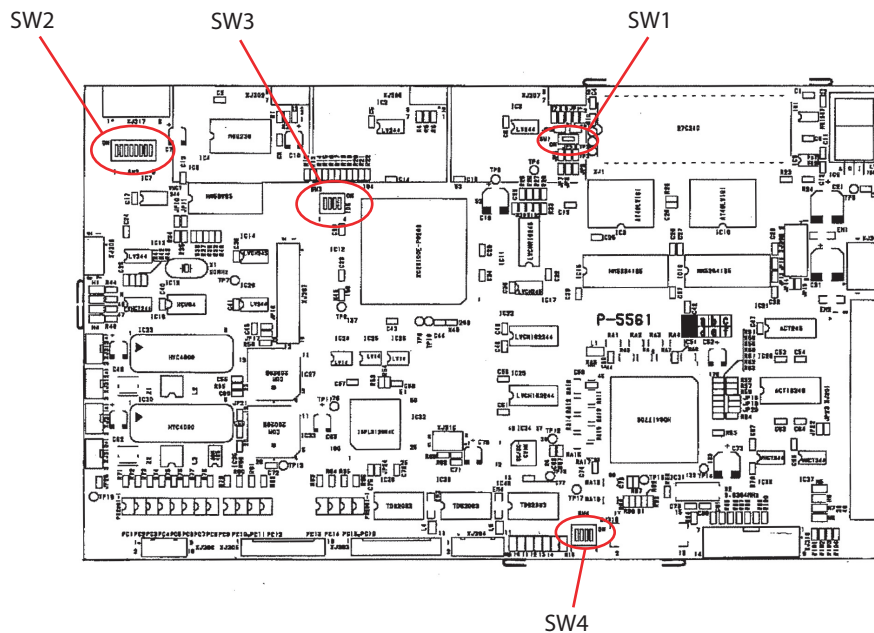


Fig. 13-41

Functions of the board

1. Processing of combination weighing.
2. Sending of control command to each unit based on the combination weighing result.
3. Sending and receiving of various data to and from the DMU board.

DIP-SW settings

Table 13-33

DIP SW	Setting
SW1	OFF
SW2-1 to 8	OFF * Switches may be set to ON depending on the specification. Check the setting when replacing the board.
SW3-1	ON
SW3-2	OFF
SW3-3	ON
SW3-4	OFF
SW4-1	OFF
SW4-2	OFF
SW4-3	ON
SW4-4	OFF

13.17.3 DMU Board (P-5562*)

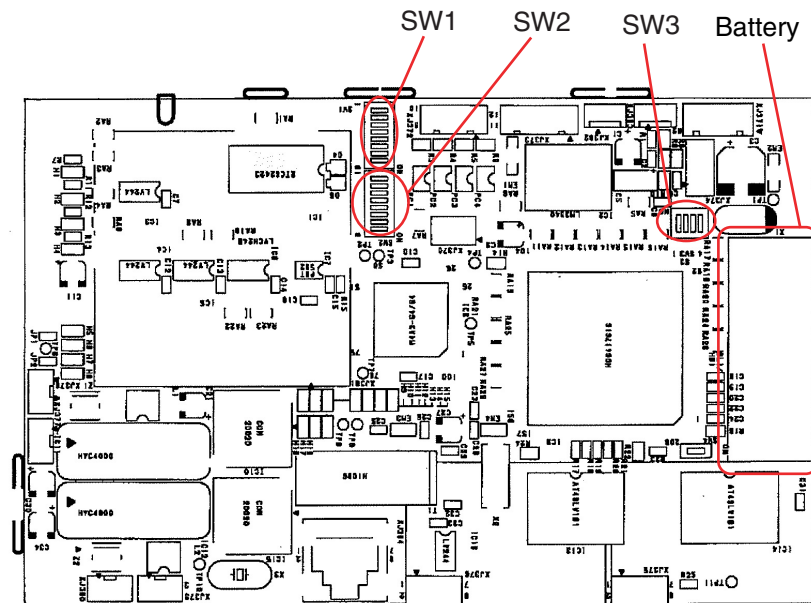


Fig. 13-42

Functions of the board

1. Data processing.
2. Storing of total data and parameters.
3. Communication with the remote control BOX unit (RCU board) by a high-speed communication line.

DIP-SW settings

Table 13-34

DIP SW	Function	Default Setting
SW1-1	ON:Models with two distributed cameras OFF:Models with one distributed camera	OFF
SW1-2	ON:Models with no camera OFF:Models with camera	OFF
SW1-3 to SW1-8	Not used	OFF
SW2-1 to SW2-5	Not used	OFF
SW2-6	ON:Clears SRAM when initializing memory OFF:Does not clear SRAM when initializing memory	OFF
SW2-7	ON:Does not output FIFO at DMU application OFF:Outputs FIFO at DMU application	OFF
SW2-8	ON:Does not output boot loader message OFF:Outputs boot loader message	OFF
SW3-1	ON:Boot loader mode OFF:Application execution mode	OFF

Table 13-34

DIP SW	Function	Default Setting
SW3-2	Fix to OFF and should no be turned ON.	OFF
SW3-3	ON:Writing data to flash memory is prohibited. OFF:Writing data to flash memory is allowed.	OFF
SW3-4	Fix to OFF and should no be turned ON.	OFF

CAUTION

- Refer to "10.3.8 Replacement of Memory Backup Battery" for replacing battery.

13.17.4 485 Board (P-5457*)

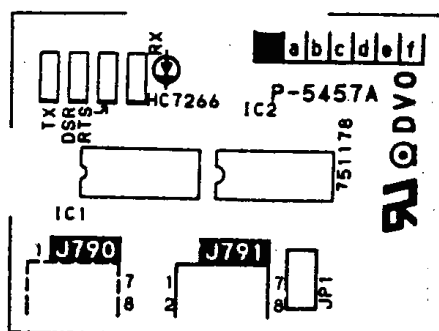


Fig. 13-43

Functions of the board

1. Conversion of data communication level between WCU and DUC.

13.18 PS FEEDER Unit

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

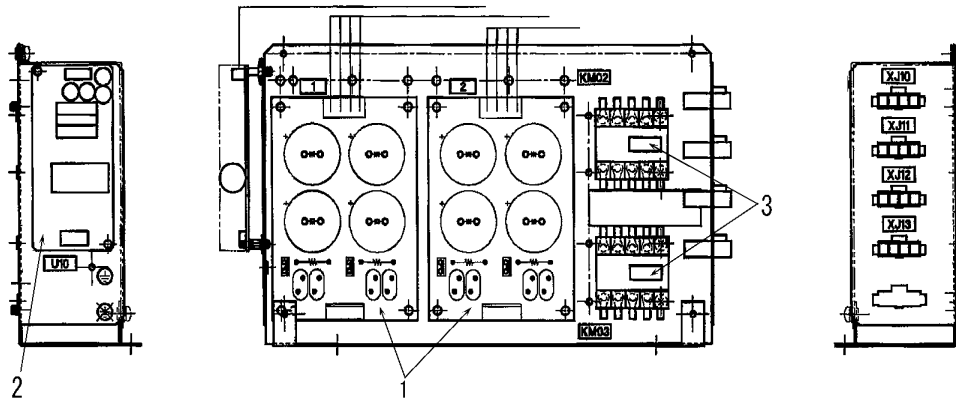


Fig. 13-44

Table 13-35

No.	Name
1	FPS board (P-5580*)
2	Switching power supply (DC12 V)
3	Electromagnetic contactor

NOTE

- The figure is an example of CCW-RV-214W*.

13.18.1 FPS Board (P-5580*)

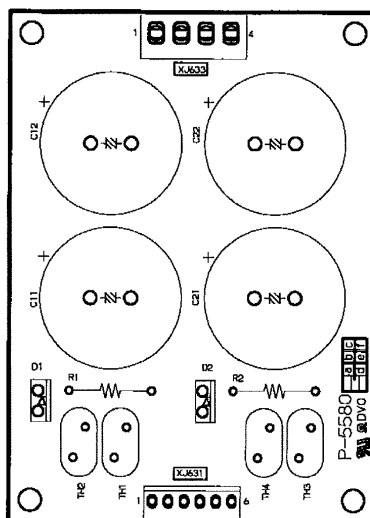


Fig. 13-45

Functions of the board

1. Converts AC power to DC power (Feeder power).

13.19 DUC Unit (CCW-RV-214W-S/05, 08)

⚠ DANGER

- All repairing the device or replacing part can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

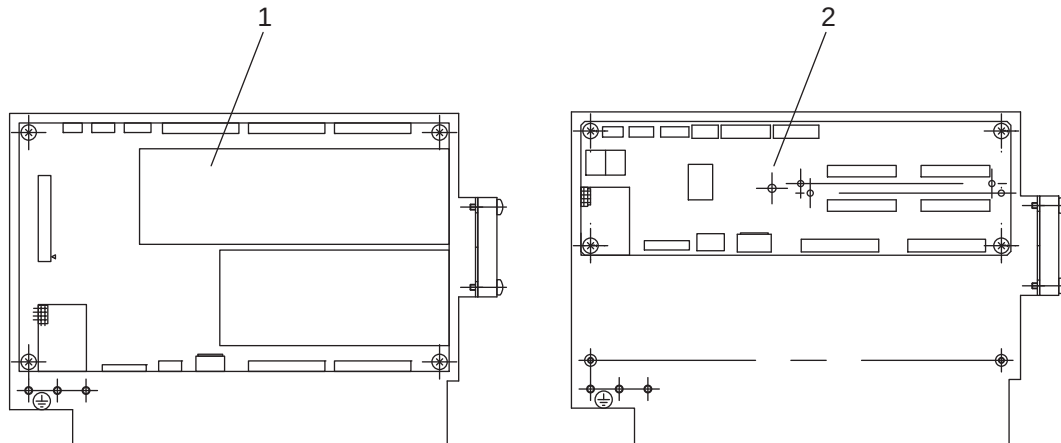


Fig. 13-46

Table 13-36

No.	Name
1	DUC board (P-5524*)
2	DUC board (P-5579*)

DIP-SW settings

NOTE

- When replacing the weigh drive unit and DUC board, perform the head setting by DIP-SW.
- The following setting is an example of CCW-RV-214W*/05, 08.

Table 13-37

DIP SW	1st (P-5524*) (for PH1-5, WH1-5)	2nd (P-5579*) (for PH6-10, WH6-10)	3rd (P-5524*) (for PH11-14, WH11-14)	4th (P-5579*) (for TH, DTH1, 2, RS)
SW1-1	ON	OFF	ON	ON
SW1-2	OFF	ON	ON	ON
SW1-3	OFF	OFF	OFF	ON
SW1-4	OFF	OFF	OFF	ON

Table 13-37

DIP SW	1st (P-5524*) (for PH1-5, WH1-5)	2nd (P-5579*) (for PH6-10, WH6-10)	3rd (P-5524*) (for PH11-14, WH11-14)	4th (P-5579*) (for TH, DTH1, 2, RS)
SW2-1	—	—	—	OFF
SW2-2	—	—	—	OFF
SW2-3	—	—	—	OFF
SW2-4	—	—	—	OFF

13.19.1 DUC Board (P-5524*)

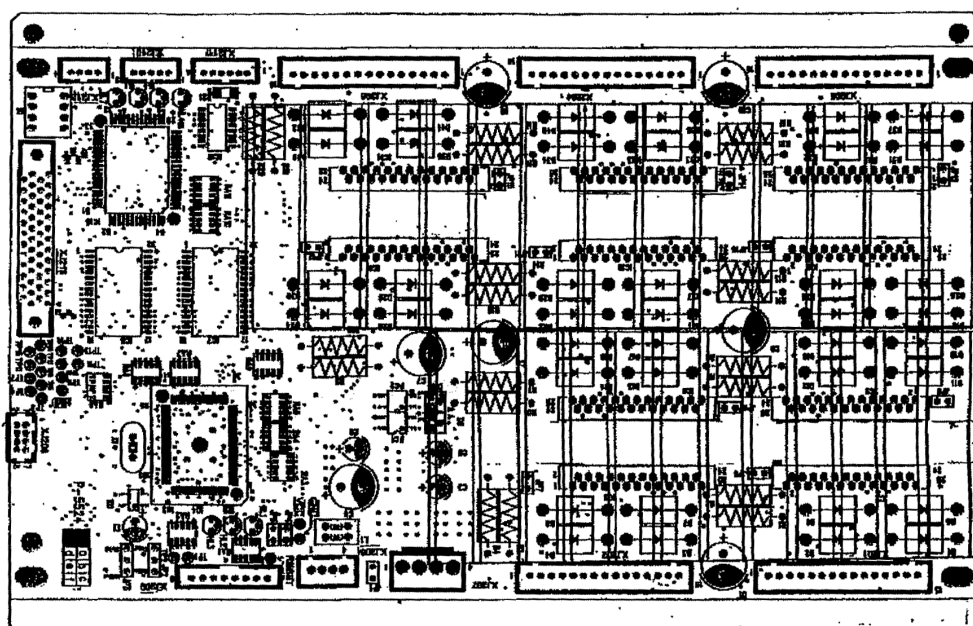


Fig. 13-47

Functions of the board

1. Drive control of PH and WH.

13.19.2 DUC Board (P-5579*)

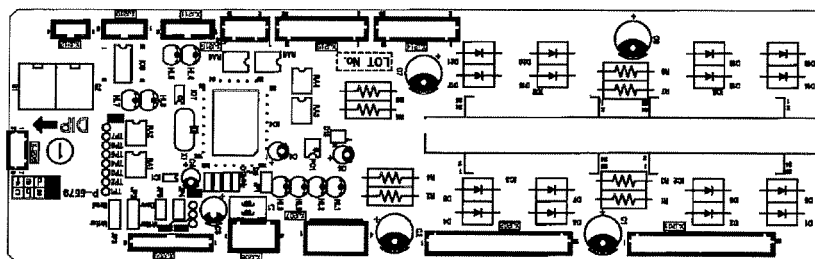


Fig. 13-48

Functions of the board

1. Drive control of TH, DTH, RS, etc.

13.20 Connection Diagram

13.20.1 TOTAL DIAGRAM

- | | |
|-------------------------------------|------------------------------|
| 1. CCW-RV-214W-S/15, 20, 30, 50, 70 | ▶ 143-6328-05
143-6329-09 |
| 2. CCW-RV-210W-S/15, 20, 30, 50, 70 | ▶ 143-6330-03
143-6331-07 |

13.20.2 BLOCK DIAGRAM

NOTE

- The drawing is targeted for CCW-RV-214W-S/15, 20, 30, 50, 70.

- | | |
|---|------------------------------|
| 1. REMOTE CONTROL UNIT | ▶ 144-8344-00 |
| 2. RELAY UNIT | ▶ 107-2454-00 |
| 3. AFV UNIT | ▶ 107-2452-02 |
| 4. ELECTRICAL UNIT | ▶ 107-2453-06 |
| 5. WEIGH DRIVE UNIT | ▶ 107-2455-03
107-2456-07 |
| 6. CAL UNIT | ▶ 105-0490-02
105-3665-08 |
| 7. POWER SUPPLY UNIT [for CCW-RV-214W-S/15, 20, 30, 50, 70] | |
| • AC200-240 V (standard) | ▶ 139-0945-06 |
| • AC200-240 V (EU) | ▶ 139-0003-08 |
| • AC380-480 V (common) | ▶ 139-6849-03 |

13.21 Connection Diagram (CCW-RV-214W-S/05, 08)

13.21.1 TOTAL DIAGRAM

1. CCW-RV-214W*/05,08 ▶ 143-6332-01
143-6333-04

13.21.2 BLOCK DIAGRAM

NOTE

- The drawing is targeted for CCW-RV-214W*/05, 08.
-

1. POWER SUPPLY UNIT

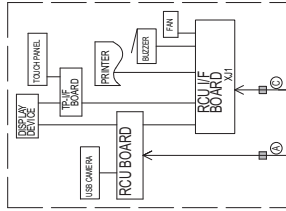
AC200-240 V (standard)

▶ 139-0933-01

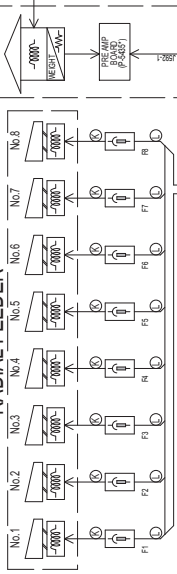
AC200-240 V (EU)

▶ 139-0935-09

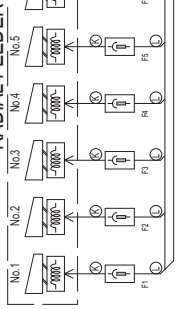
REMOTE CONTROL BOX UNIT



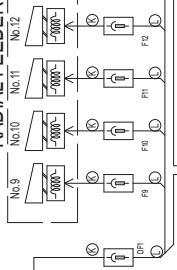
DISPERSION FEEDER



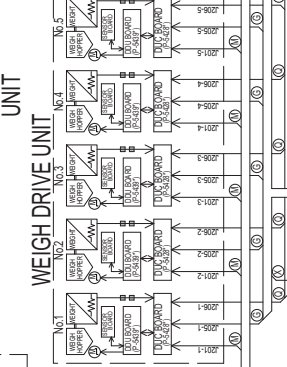
RADIAL FEEDER



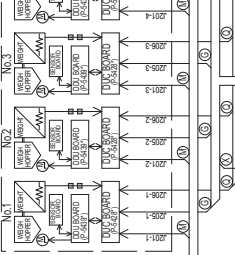
RADIAL FEEDER



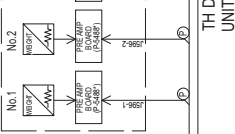
ELECTRICAL UNIT



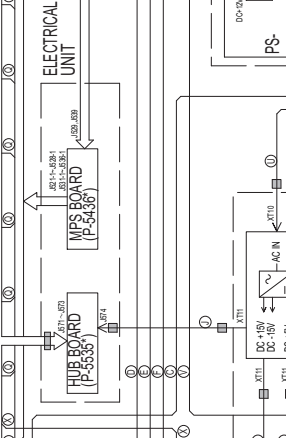
WEIGH DRIVE UNIT



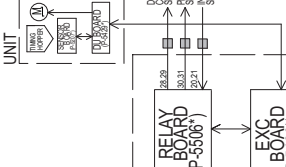
AFV UNIT



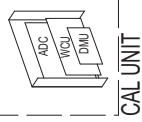
ELECTRICAL UNIT



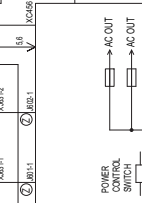
TH DRIVE UNIT



CAL UNIT



PS UNIT



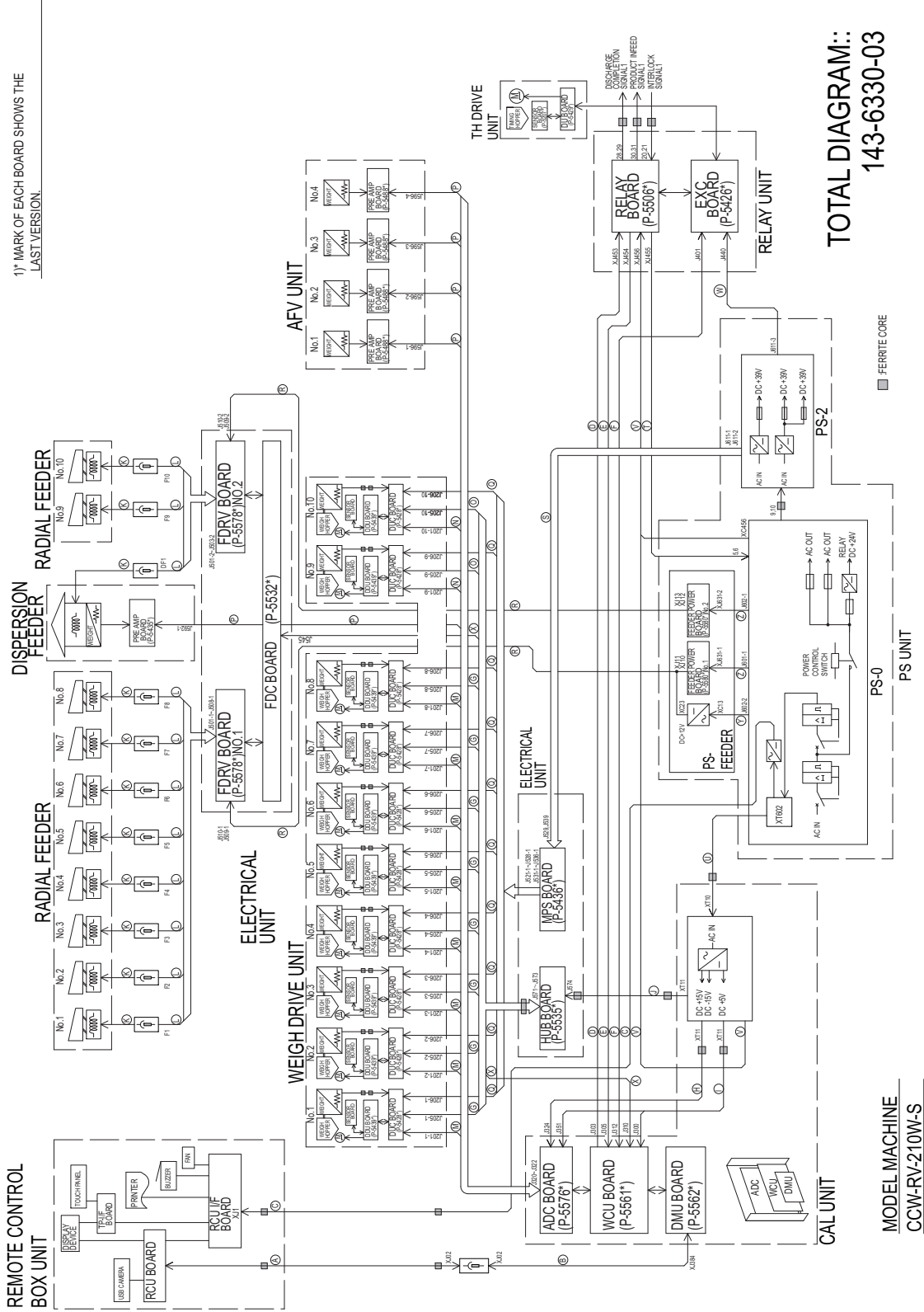
RELAY UNIT



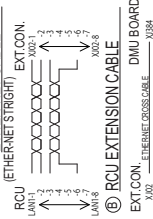
MODEL MACHINE
CCW-RV-214W-S

TOTAL DIAGRAM::
143-6328-05

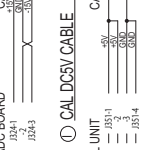
1)* MARK OF EACH BOARD SHOWS THE
LATEST VERSION.



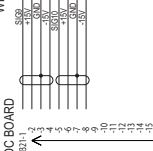
① RCU COMMUNICATION ETHERNET STRIGHT



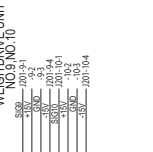
② RCU EXTENSION CABLE



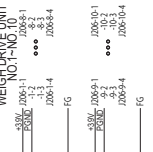
③ CAL DC15V CABLE



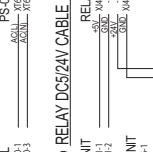
④ CAL DC5V CABLE



⑤ HUB DC5V CABLE



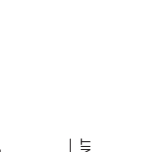
⑥ RCU PS CABLE



⑦ EXG-OUT CABLE



⑧ EXG-IN CABLE



⑨ EXG COMMUNICATION CABLE



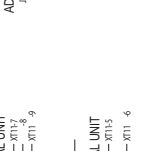
⑩ PS-CAL PS CABLE



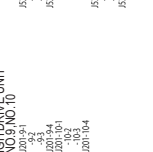
⑪ RELAY DC5V/24V CABLE



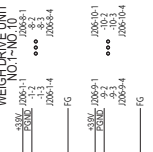
⑫ PS-0 UNIT



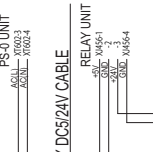
⑬ RELAY DC39V CABLE



⑭ DUC COMMUNICATION CABLE



⑮ MOTOR PS CABLE



⑯ FEEDER POWER CABLE



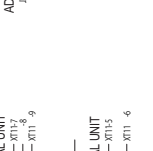
⑰ DC FUSE CABLE



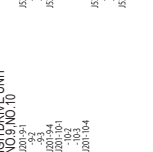
⑱ PS FEEDER AC CABLE



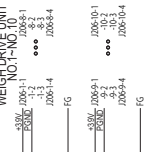
⑲ AC FUSE CABLE



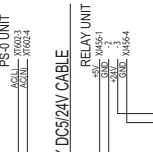
⑳ POWER CONTROL CABLE



㉑ WEIGHT SIG CABLE



㉒ DUC COMMUNICATION CABLE



㉓ WEIGHT SIG CABLE



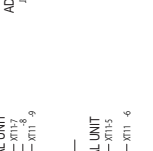
㉔ WEIGHT SIG CABLE



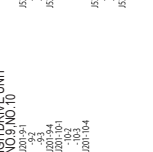
㉕ WEIGHT SIG CABLE



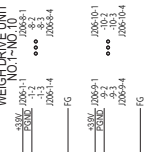
㉖ WEIGHT SIG CABLE



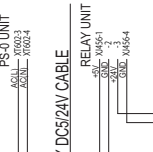
㉗ WEIGHT SIG CABLE



㉘ WEIGHT SIG CABLE



㉙ WEIGHT SIG CABLE



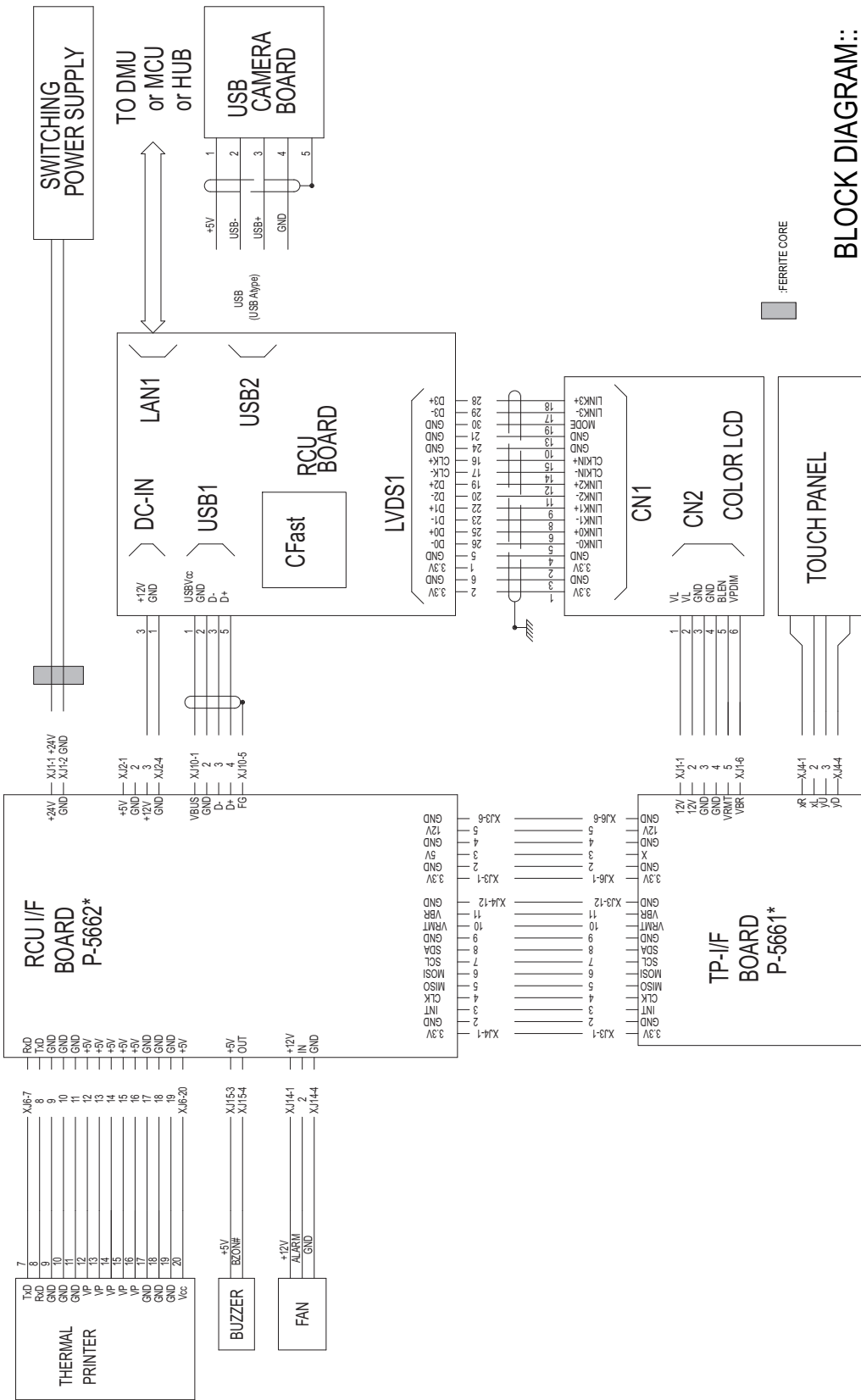
㉚ WEIGHT SIG CABLE



TOTAL DIAGRAM::
143-6331-07

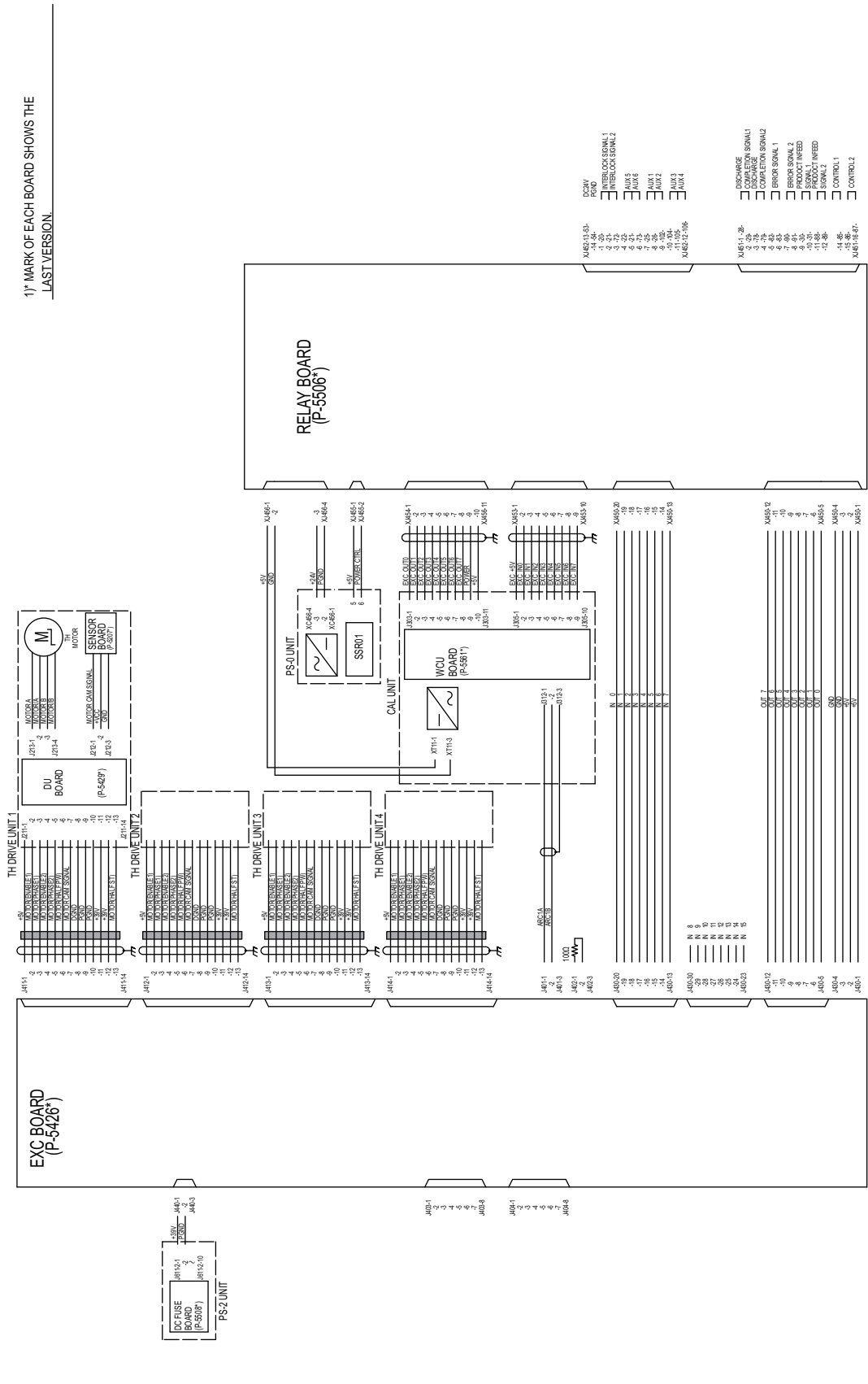
MODEL MACHINE
CCW-RV-210W-S

NOTE) "xxx".Alphanumerical code which corresponds
to the latest version of the PWB.



REMOTE CONTROL UNIT

BLOCK DIAGRAM::
144-8344-00



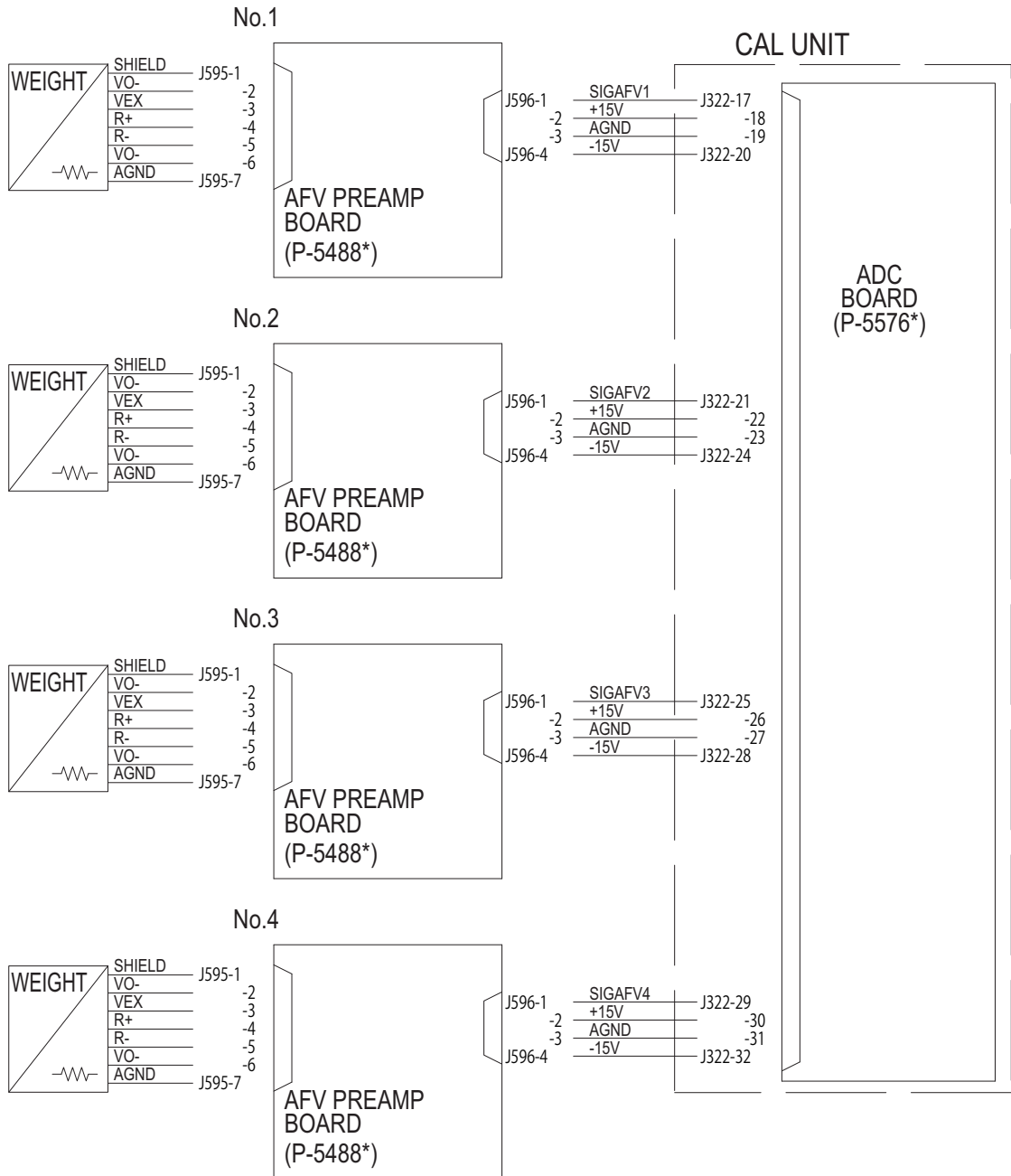
RELAY UNIT

MODEL MACHINE:
CCW-RIRV-2** SERIES
CCW-RSIRVE-2** SERIES

BLOCK DIAGRAM::
107-2454-00

■ : FERRITE CORE

1)* MARK OF EACH BOARD SHOWS THE
LATEST VERSION.



AFV UNIT

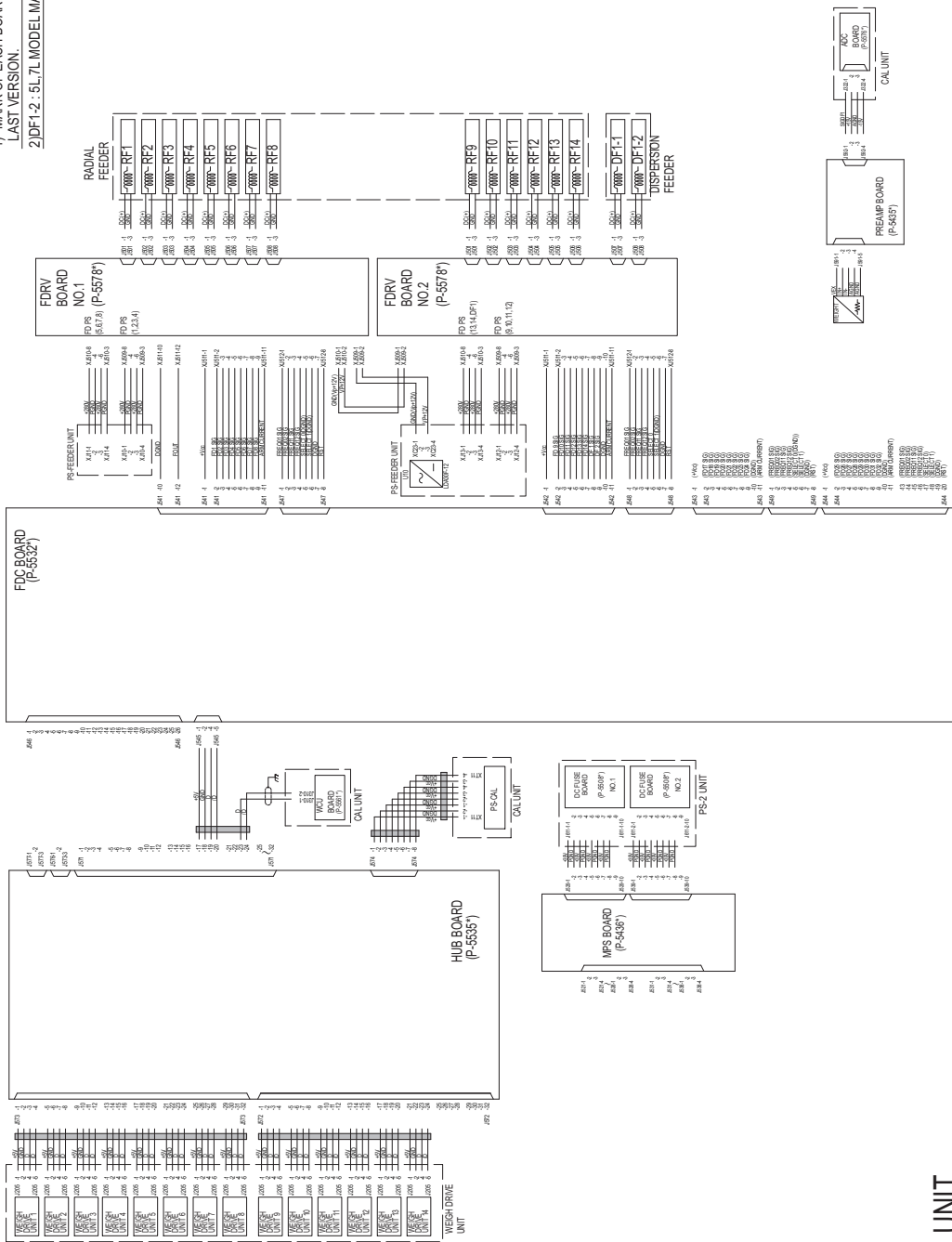
MODEL MACHINE:

CCW-R SERIES

CCW-RV SERIES

BLOCK DIAGRAM::
107-2452-02

1)* MARK OF EACH BOARD SHOWS THE
LAST VERSION.
2)DF1-2 : 5L, 7L MODEL MACHINE.



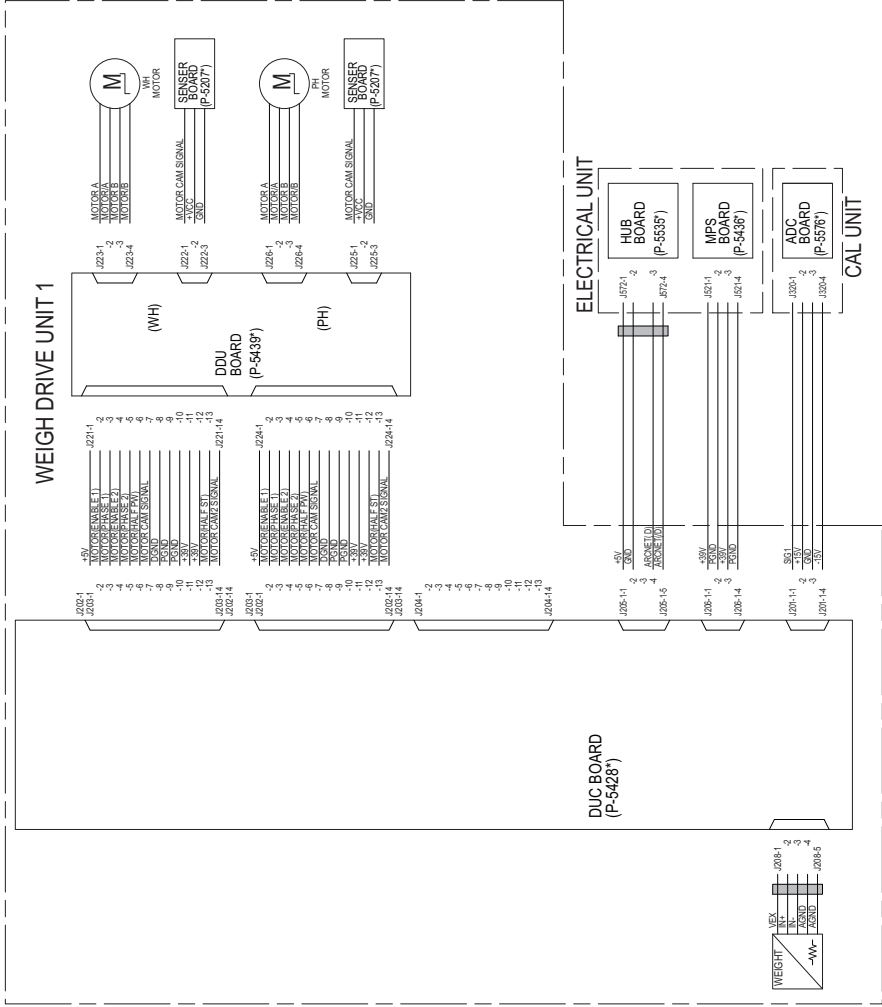
ELECTRICAL UNIT

MODEL MACHINE:
CCW-R-214W-S
CCW-RV-214W-S

BLOCK DIAGRAM:: 107-2453-06

■ FERRITE CORE

1)* MARK OF EACH BOARD SHOWS THE
LATEST VERSION.



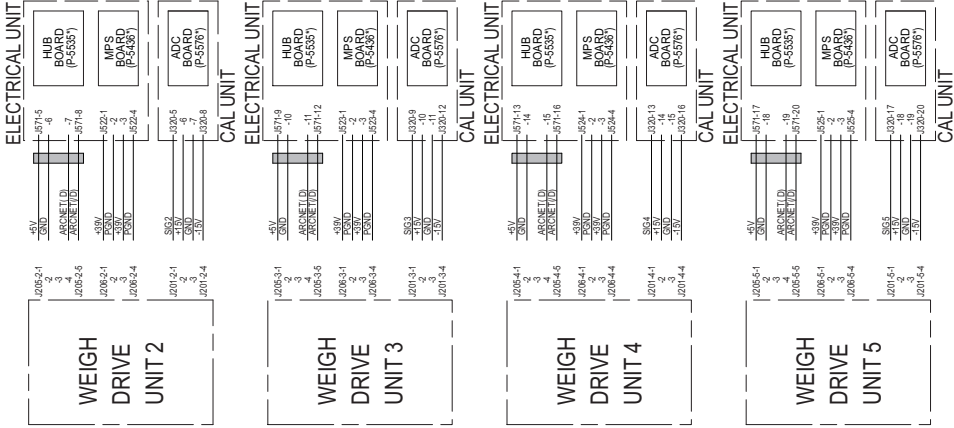
WEIGH DRIVE UNIT 1/2

MODEL MACHINE:

CCW-R-2** SERIES (2 MOTOR)

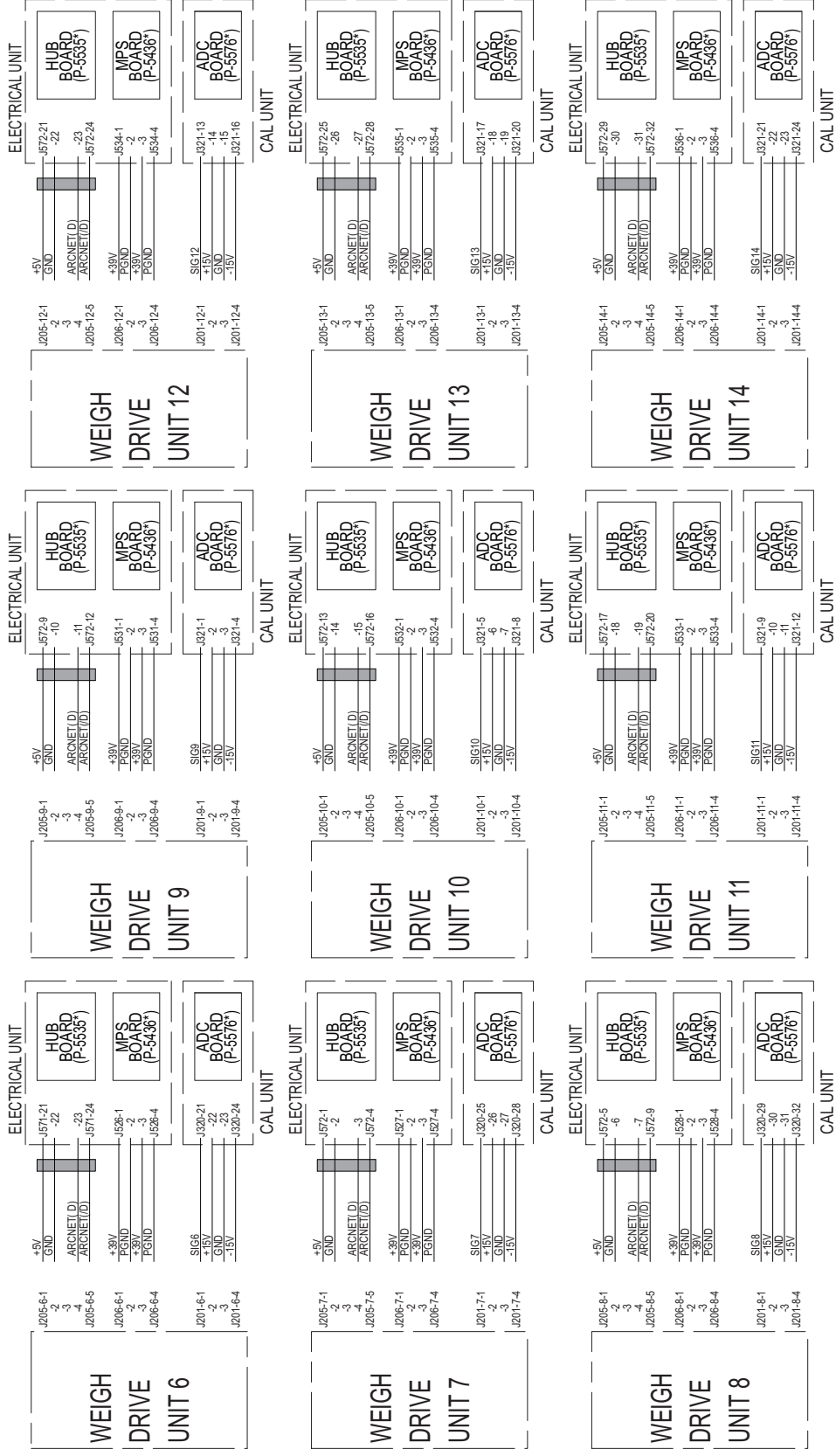
CCW-RV-2** SERIES (2 MOTOR)

■ FERRITE CORE



BLOCK DIAGRAM:: 107-2455-03

1)* MARK OF EACH BOARD SHOWS THE
LATEST VERSION.



WEIGH DRIVE UNIT 2/2

MODEL MACHINE:

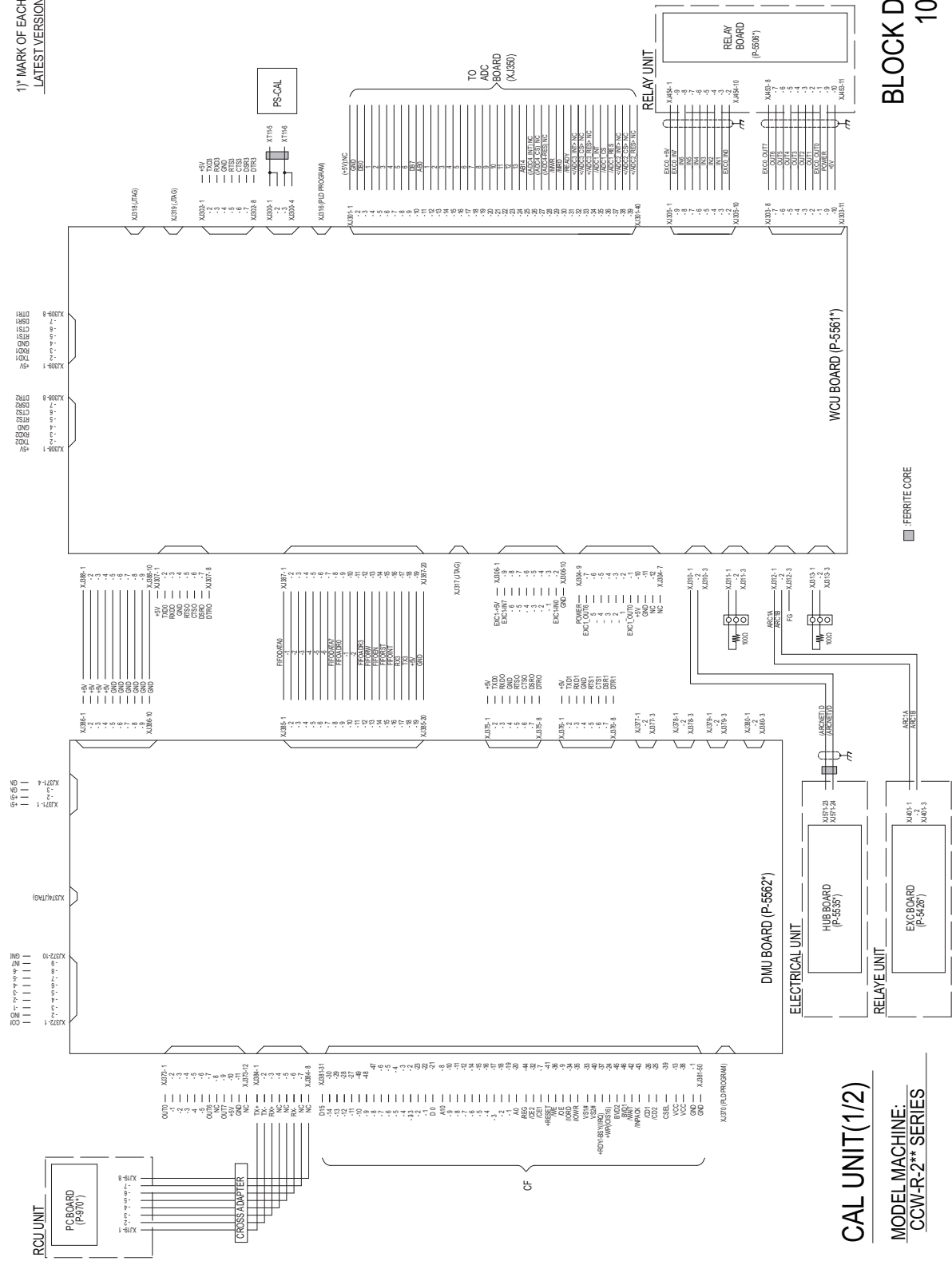
CCW-R-2** SERIES (2 MOTOR)

CCW-RV-2** SERIES (2 MOTOR)

■ FERRITE CORE

BLOCK DIAGRAM::
107-2456-07

1)* MARK OF EACH BOARD SHOWS THE
LATEST VERSION.

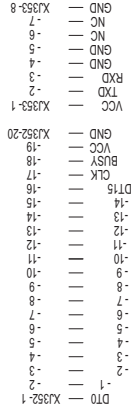


CAL UNIT(1/2)

MODEL MACHINE:
CCW-R-2** SERIES

BLOCK DIAGRAM::
105-0490-02

Diagram illustrating the PS-CAL system architecture. The PS-CAL system is connected to three external systems: XT11-7, XT11-9, and XT11-8. These systems are connected to a common line, which then connects to a vertical line leading to a horizontal line at the top.



MODEL MACHINE:

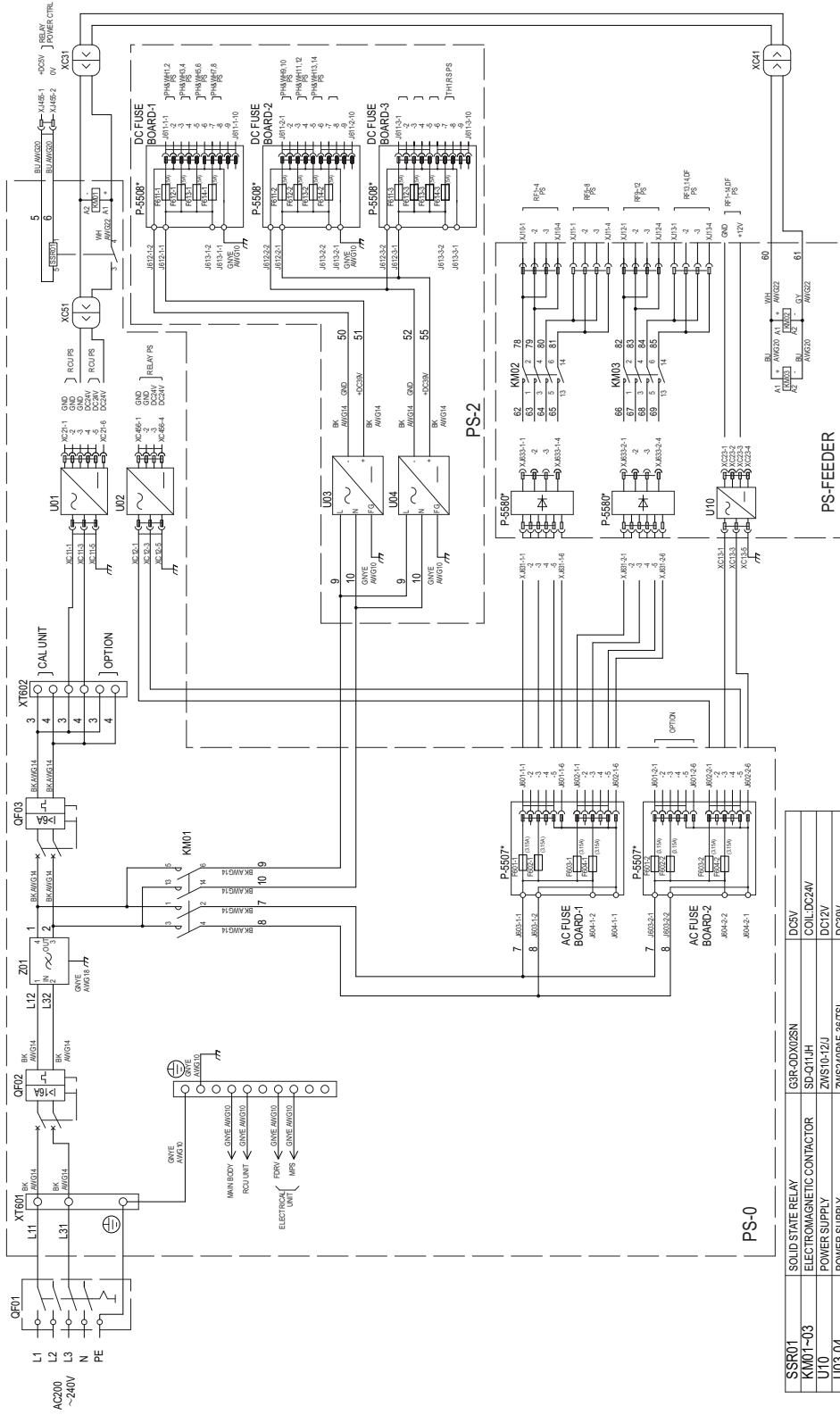
CCW-RV-2** SERIES

CCW-RV-2** SERIES

■ FERRITE CORE

BLOCK DIAGRAM::
105-3665-08

1" MARK OF EACH BOARD SHOWS THE
LAST VERSION.



CCW-RV-214W-S/15,20,30,50,70

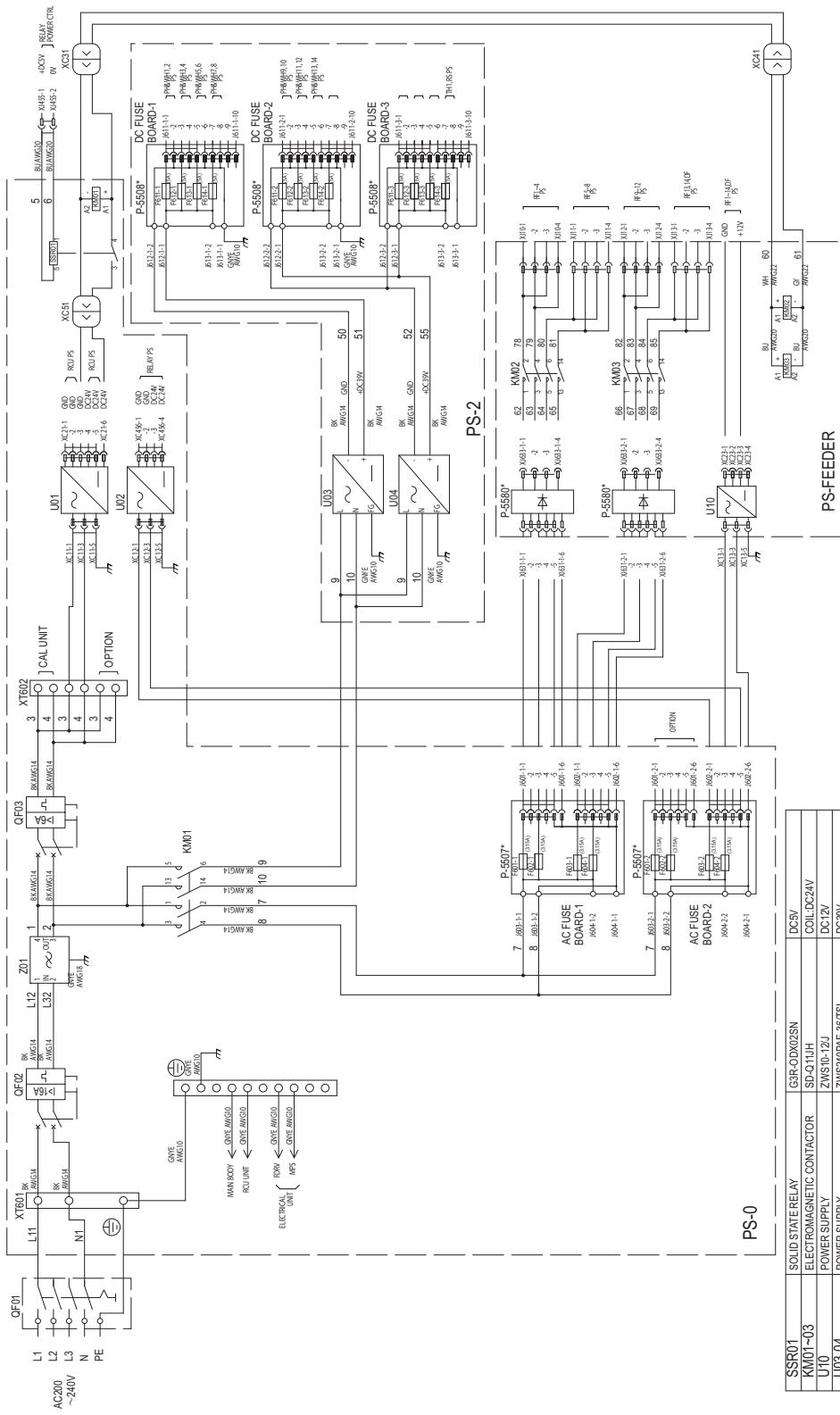
CCW-R-214W-S/15,20,30,50,70

AC200-240V(STD)

BLOCK DIAGRAM::
139-0945-06

Item Designation	PART NAME	TYPE	REMARKS/RATING
SSR01	SOLID STATE RELAY	GJR-ODX02SN	DC5V
XM01-03	ELECTROMAGNETIC CONTACTOR	SC-QTJH	COLL.DC24V
U10	POWER SUPPLY	ZWS10-12J	DC12V
U03,04	POWER SUPPLY	ZWS240PAF-36TSl	DC39V
U02	POWER SUPPLY	ZWS30-24J	DC24V
U01	POWER SUPPLY	ZWS75AF-24J	DC24V
Z01	NOISE FILTER	ZAG220-11S	AC280V 20A
QF03	CIRCUIT PROTECTOR	FAZ-S802	6A
QF02	CIRCUIT BREAKER	FAZ-S162	16A
QF01	EMERGENCY SWITCH	P1-25EASVB	3PAC300V/25A

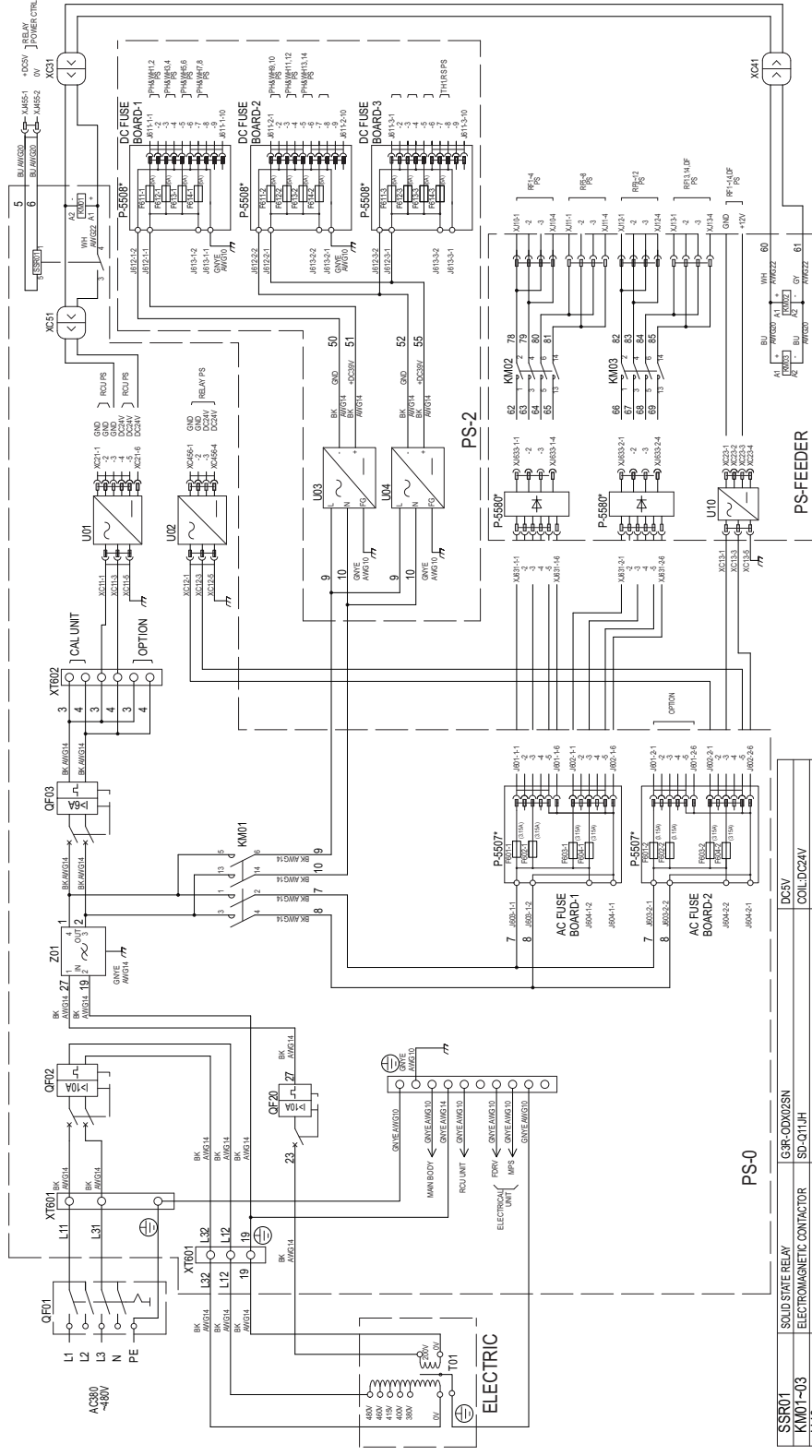
1)* MARK OF EACH BOARD SHOWS THE
LAST VERSION.



CCW-RV-214W-S/15,20,30,50,70
CCW-R-214W-S/15,20,30,50,70
AC200~240V(EU)

BLOCK DIAGRAM:::
139-0003-08

1)* MARK OF EACH BOARD SHOWS THE LAST VERSION.



CCW-RV-214W-S/15,20,30,50,70
CCW-R-214W-S/15,20,30,50,70
AC380~480V(EU)

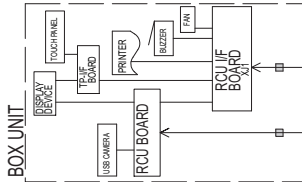
BLOCK DIAGRAM::
139-6849-03

SSR01	SOLID STATE RELAY	GJR-ODX02SN	DC5V
KM01-03	ELECTROMAGNETIC CONTACTOR	SD-Q111H	COIL:DC24V
U10	POWER SUPPLY	ZWS10-12JJ	DC12V
U03_04	POWER SUPPLY	ZWS240PAF-36TISI	DC36V
U02	POWER SUPPLY	ZWS30-24JJ	DC24V
U01	POWER SUPPLY	LENFSE-24	DC24V
Z01	NOISE FILTER	ZAG2220-1HS	DC260V/20A
T01	TRANSFORMER	ZS1530-30 Z1ER6X60(24W S5010)	500V/0.041546/0.0486V
QF20	CIRCUIT BREAKER	FAZ-S10	10A
QF02	CIRCUIT PROTECTOR	FAZ-S602	6A
QF03	CIRCUIT BREAKER	PKZM0-10-T	10A
QF01	EMERGENCY SWITCH	KG0120433KSTIV	3P+NACT50 25A
Item Designation	PART NAME	TYPE	REMARKS/RATING

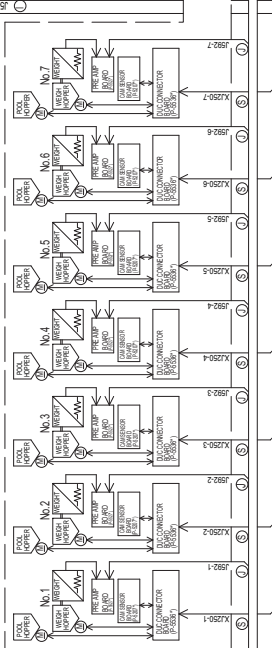
DUC DIPSW SETTING

	DUC BOARD1	DUC BOARD2	DUC BOARD3	DUC BOARD4
DPSW1	1 ON	2 OFF	3 ON	4 ON
	2 OFF	3 ON	4 OFF	1 ON
DPSW2	1 OFF	2 OFF	3 OFF	4 OFF
	2 OFF	3 OFF	4 OFF	1 OFF
	2 OFF	3 OFF	4 OFF	1 OFF
	2 OFF	3 OFF	4 OFF	1 OFF

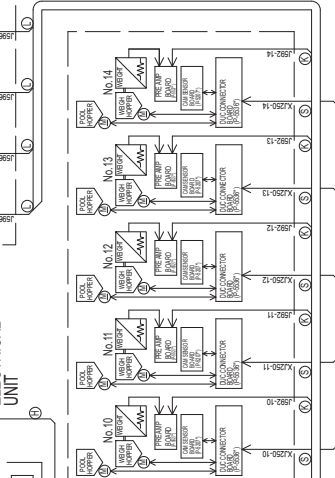
REMOTE CONTROL BOX UNIT



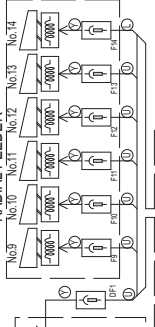
WEIGH DRIVE UNIT



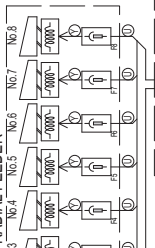
ELECTRICAL UNIT



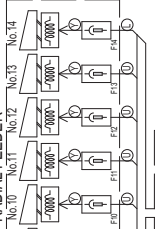
DISPERSION FEEDER



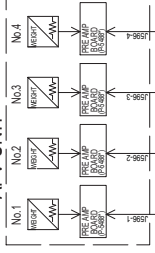
RADIAL FEEDER



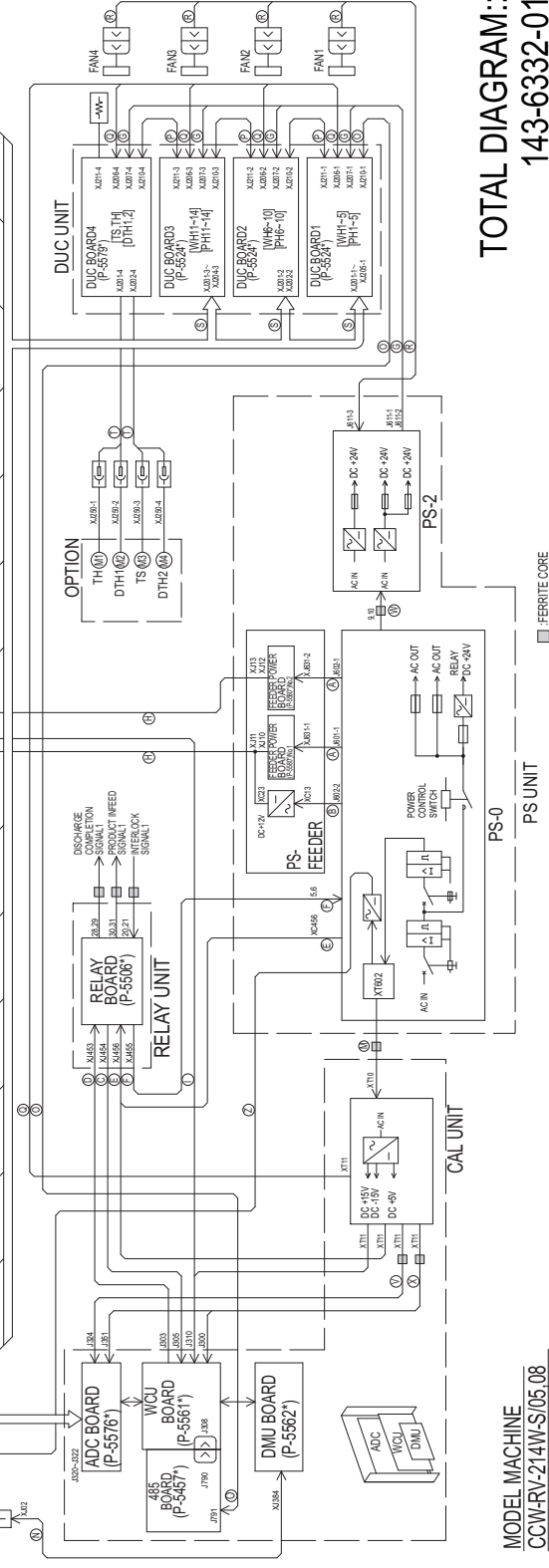
RADIAL FEEDER



AFV UNIT



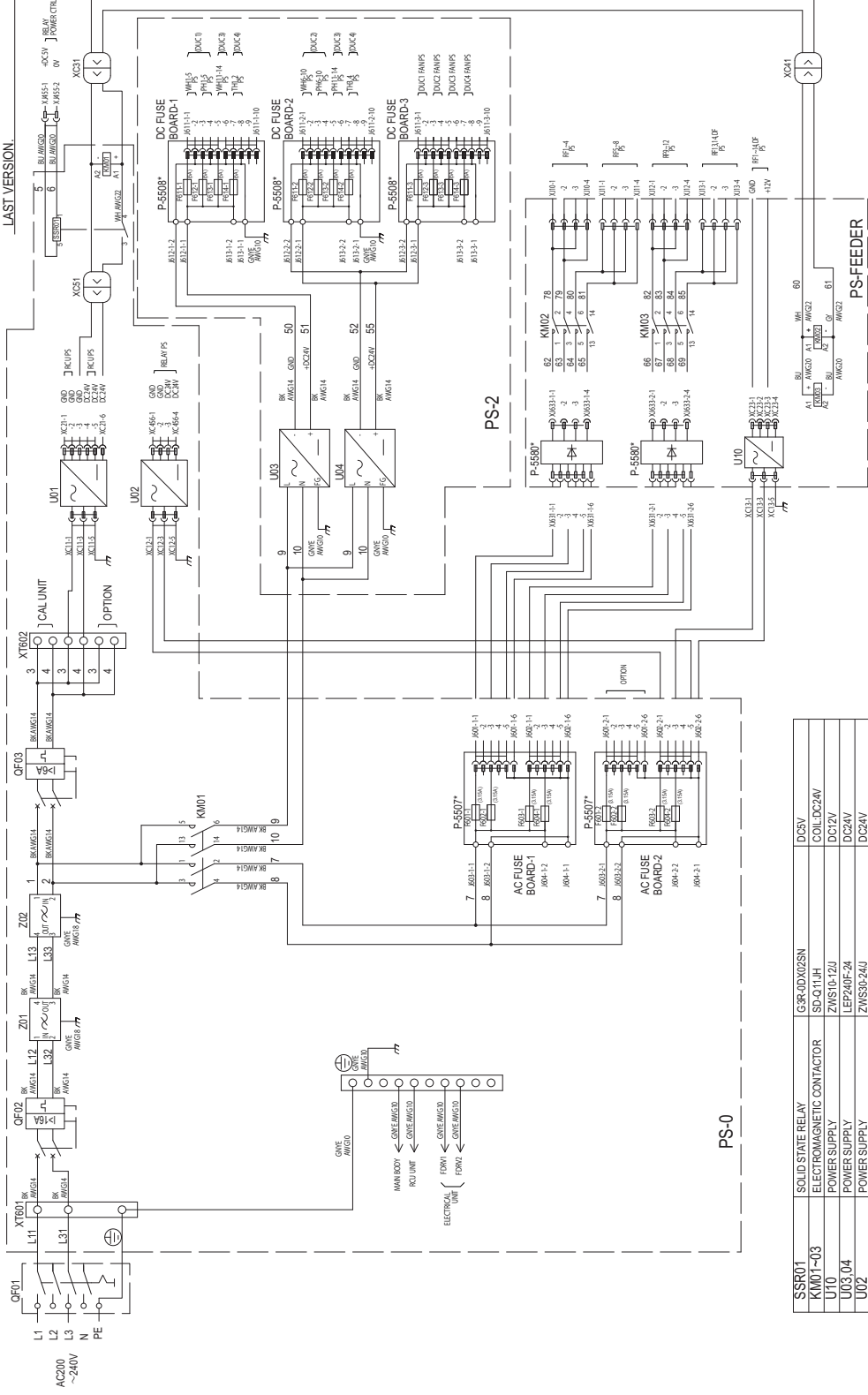
1* MARK OF EACH BOARD SHOWS THE LATEST VERSION.



TOTAL DIAGRAM::
143-6332-01

MODEL MACHINE
CCWR-RV-214W-S05_08

1)* MARK OF EACH BOARD SHOWS THE
LAST VERSION.



CCW-RV-214W-S/05.08

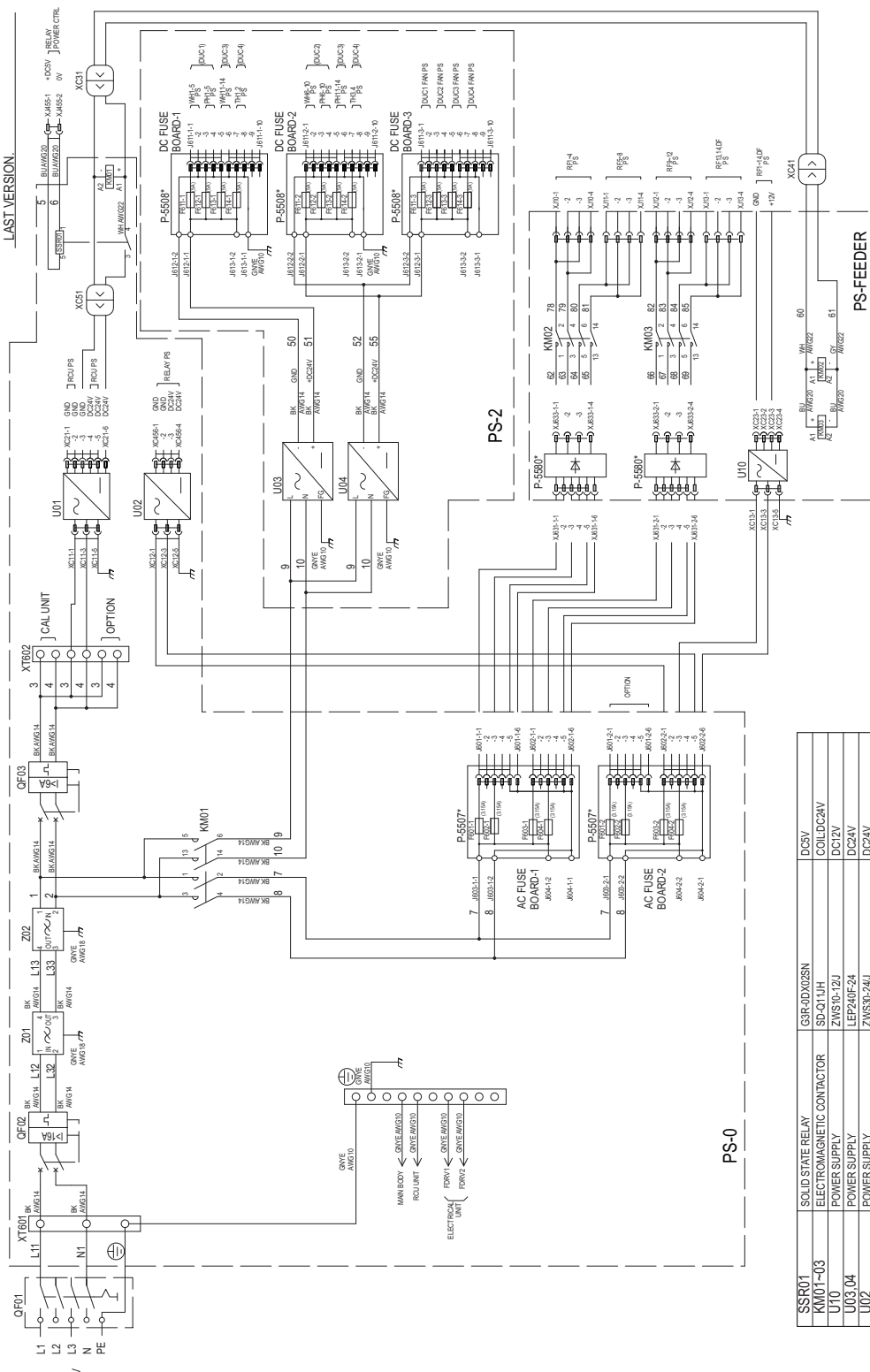
CCW-R-214W-S/05.08

AC200~240V(STD)

BLOCK DIAGRAM::
139-0933-01

ITEM DESIGNATION	PART NAME	TYPE	REMARKS/RATING
SSR01	SOLID STATE RELAY	GJR4DX02SN	DC5V
KM01~03	ELECTROMAGNETIC CONTACTOR	SD-Q11JH	COIL:DC24V
U10	POWER SUPPLY	ZWS10-12J	DC12V
U03.04	POWER SUPPLY	LEP240F-24	DC24V
U02	POWER SUPPLY	ZWS30-24J	DC24V
U01	POWER SUPPLY	ZWS75AF-24J	DC24V
Z01.02	NOISE FILTER	ZAG2220-11S	AC250V 20A
QF03	CIRCUIT PROTECTOR	FAZ-S82	6A
QF02	CIRCUIT BREAKER	FAZ-S162	16A
QF01	EMERGENCY SWITCH	KG2072453K1SV	3P+N AC750V 25A

1)* MARK OF EACH BOARD SHOWS THE LAST VERSION.



CCW-RV-214W-S/05.08
CCW-R-214W-S/05.08
AC200~240V(EU)

BLOCK DIAGRAM::
139-0935-09

ITEM DESIGNATION	PART NAME	TYPE	REMARKS RATING
SSR01	SOLID STATE RELAY	GSR-0DX02SN	DC5V
KM01-03	ELECTROMAGNETIC CONTACTOR	SP-011JH	COIL:DC24V
U10	POWER SUPPLY	ZWS10-12U	DC12V
U03,04	POWER SUPPLY	LEP240F-24	DC24V
U02	POWER SUPPLY	ZWS30-24U	DC24V
U01	POWER SUPPLY	LEA75F-24	DC24V
Z01,02	NOISE FILTER	ZAG220-11S	AC250V 20A
QF03	CIRCUIT PROTECTOR	FZS-S62	6A
QF02	CIRCUIT BREAKER	FZS-S162	16A
QF01	EMERGENCY SWITCH	KG20T204/3KNS1V	3P+N AC/50V 25A

14 PERIODICAL INSPECTION TO PREVENT MIXING IN THE FOREIGN MATERIAL

Follow the procedures below (Refer to "Table 14-1") to maintain the device at its and use it for many years.

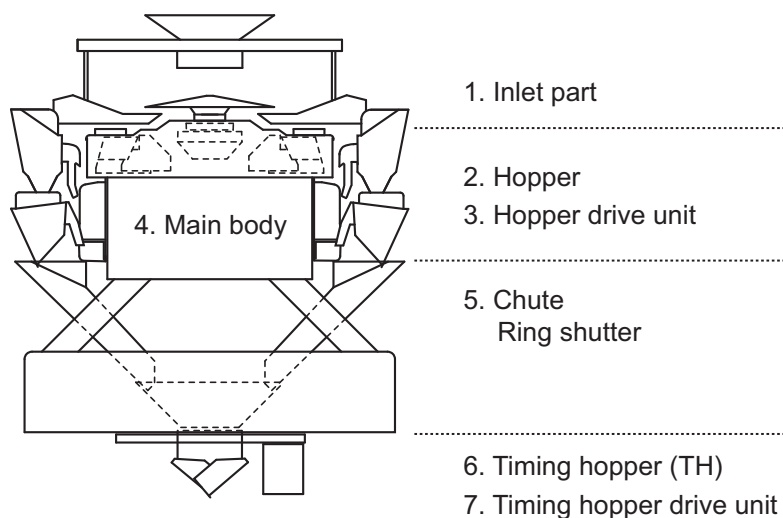


Fig. 14-1

Table 14-1

Zone	Check Result	Part and Item to be Inspected	Cycle	Exchange Term
1. Inlet Part	OK/NG	Urethane sheet It should be free from scratches, cracks, breaks, abrasions, discoloration, hardening and warping.	Every week	3 years
	OK/NG	Silicone coat (milk white) It should be free from scratches, cracks, breaks, abrasions, discoloration, hardening and warping.	Every week	3 years
	OK/NG	Coating (Teflon®, Urethane or Plating) It should be free from scratches and peeling.	Every day	Depends on the conditions of use
	OK/NG	Drive chain of shutter at leading edge of trough. It should be free from rust and dirt.	Every week	5 years

Table 14-1

Zone	Check Result	Part and Item to be Inspected	Cycle	Exchange Term
2.Hopper	OK/NG	Urethane sheet It should be free from scratches, cracks, breaks, abrasions, discoloration, hardening and warping.	Every week	3 years
	OK/NG	Coating (Teflon®, Urethane or Plating) It should be free from scratches and peeling.	Every day	Depends on the conditions of use
	OK/NG	Mesh (resin or stainless steel) It should be free from scratches, fraying, discoloration, hardening and warping.	Every day	1 year
	OK/NG	Bearing	Every week	When play develops
	OK/NG	Bushes (support points for shaft) It should be free from scratches, cracks, breaks, abrasions, discoloration, hardening and warping.	Every week	5 years
3.Hopper Drive	OK/NG	Seal washer (flat washer with rubber coating)	When removed	When removed
	OK/NG	Packing It should be free from discoloration, hardening, breaks and abrasions.	When removed	When removed after 1 year or more of use
	OK/NG	Oil seal, V ring	Every week	When removed
4.Main Body	OK/NG	Wiring fixtures It should be free from discoloration, hardening, breaks and abrasions.	Every week	5 years

Table 14-1

Zone	Check Result	Part and Item to be Inspected	Cycle	Exchange Term
5.Chute Ring shutter	OK/NG	Urethane sheet It should be free from scratches, cracks, breaks, abrasions, discoloration, hardening and warping.	Every week	3 years
	OK/NG	Coating (Teflon®, Urethane or Plating) It should be free from scratches and peeling.	Every day	Depends on the conditions of use
	OK/NG	Mesh (resin or stainless steel) It should be free from scratches, fraying, discoloration, hardening and warping.	Every day	1 year
6.Timing Hopper	OK/NG	Urethane sheet It should be free from scratches, cracks, breaks, abrasions, discoloration, hardening and warping.	Every week	3 years
	OK/NG	Coating (Teflon®, Urethane or Plating) It should be free from scratches and peeling.	Every day	Depends on the conditions of use
	OK/NG	Bushes (support points for shaft) It should be free from scratches, cracks, breaks, abrasions, discoloration, hardening and warping.	Every week	5 years
	OK/NG	Mesh (resin or stainless steel) It should be free from scratches, fraying, discoloration, hardening and warping.	Every day	1 year
	OK/NG	Bearing	Every week	When play develops
7.Timing Hopper Drive Unit	OK/NG	Seal washer (flat washer with rubber coating)	When removed	When removed
	OK/NG	Packing It should be free from discoloration, hardening, breaks and abrasions	When removed	When removed after 1 year or more of use
	OK/NG	Oil seal, V ring	Every week	When removed

